

# Causal Inference in Economics and Marketing

## Section 07: Difference-in-Differences and Its Variants

Weiheng Zhang

September 05, 2026

### Contents

|          |                                                                            |           |
|----------|----------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>The problem</b>                                                         | <b>2</b>  |
| 1.1      | Same question, a new source of variation . . . . .                         | 2         |
| 1.2      | One experiment, the classic regression breaks three ways . . . . .         | 3         |
| 1.3      | Reading the results . . . . .                                              | 5         |
| 1.4      | Roadmap . . . . .                                                          | 5         |
| <b>2</b> | <b>Just enough framework</b>                                               | <b>5</b>  |
| 2.1      | Potential outcomes indexed by adoption time . . . . .                      | 5         |
| 2.2      | Three assumptions . . . . .                                                | 6         |
| 2.3      | Identification of the 2x2 ATT: skeleton . . . . .                          | 6         |
| 2.4      | From two periods to long differences . . . . .                             | 7         |
| 2.5      | What parallel trends really assumes . . . . .                              | 7         |
| 2.6      | Inference from influence functions . . . . .                               | 8         |
| <b>3</b> | <b>Covariates: three faces of conditional parallel trends</b>              | <b>8</b>  |
| 3.1      | Why covariates . . . . .                                                   | 8         |
| 3.2      | The tempting regression and why it fails . . . . .                         | 9         |
| 3.3      | Outcome regression, IPW, doubly robust . . . . .                           | 9         |
| 3.4      | DR-DiD is the ATT score of Section 04 on the differenced outcome . . . . . | 10        |
| 3.5      | Machine-learning nuisances and cross-fitting . . . . .                     | 10        |
| 3.6      | Repeated cross-sections . . . . .                                          | 11        |
| <b>4</b> | <b>Dynamics with two groups</b>                                            | <b>11</b> |
| 4.1      | ATT(g,t) as an event study . . . . .                                       | 11        |
| 4.2      | Anticipation . . . . .                                                     | 12        |
| 4.3      | TWFE event-study regressions . . . . .                                     | 13        |
| 4.4      | Simultaneous confidence bands . . . . .                                    | 13        |
| <b>5</b> | <b>Staggered adoption: what goes wrong</b>                                 | <b>13</b> |
| 5.1      | Static TWFE as a weighted average . . . . .                                | 13        |
| 5.2      | Negative weights . . . . .                                                 | 14        |
| 5.3      | Dynamic TWFE and contamination . . . . .                                   | 14        |
| 5.4      | What is not the problem . . . . .                                          | 15        |
| <b>6</b> | <b>Staggered adoption: reliable estimators</b>                             | <b>16</b> |
| 6.1      | Identification, aggregation, inference as three steps . . . . .            | 16        |
| 6.2      | The building block ATT(g,t) . . . . .                                      | 16        |
| 6.3      | Aggregations . . . . .                                                     | 17        |
| 6.4      | Sun and Abraham . . . . .                                                  | 18        |

|           |                                                           |           |
|-----------|-----------------------------------------------------------|-----------|
| 6.5       | Imputation estimators . . . . .                           | 18        |
| 6.6       | Stacked DiD and local-projections DiD . . . . .           | 18        |
| 6.7       | Comparing estimators . . . . .                            | 19        |
| 6.8       | Covariates in staggered designs . . . . .                 | 20        |
| <b>7</b>  | <b>Assessing parallel trends</b>                          | <b>21</b> |
| 7.1       | Pre-trends tests and what they cannot tell . . . . .      | 21        |
| 7.2       | Conditioning on passing the pre-test . . . . .            | 21        |
| 7.3       | Sensitivity analysis . . . . .                            | 21        |
| 7.4       | Functional-form test . . . . .                            | 22        |
| 7.5       | Placebos and design checks . . . . .                      | 22        |
| <b>8</b>  | <b>Inference</b>                                          | <b>23</b> |
| 8.1       | What level to cluster . . . . .                           | 23        |
| 8.2       | Few clusters . . . . .                                    | 23        |
| 8.3       | Design-based inference and random timing . . . . .        | 23        |
| <b>9</b>  | <b>Beyond binary absorbing treatments</b>                 | <b>24</b> |
| 9.1       | Continuous and multi-valued doses . . . . .               | 24        |
| 9.2       | Treatments that turn on and off . . . . .                 | 25        |
| 9.3       | Triple differences . . . . .                              | 25        |
| 9.4       | Nonlinear and distributional DiD . . . . .                | 26        |
| <b>10</b> | <b>Synthetic difference-in-differences</b>                | <b>26</b> |
| 10.1      | When the comparison group is not credible . . . . .       | 26        |
| 10.2      | Unit and time weights . . . . .                           | 27        |
| 10.3      | Inference . . . . .                                       | 27        |
| 10.4      | Diagnostics . . . . .                                     | 28        |
| <b>11</b> | <b>Practice</b>                                           | <b>28</b> |
| 11.1      | Replications and anchors . . . . .                        | 28        |
| 11.2      | A DiD checklist . . . . .                                 | 29        |
| <b>12</b> | <b>Extensions</b>                                         | <b>29</b> |
| <b>13</b> | <b>Appendix</b>                                           | <b>30</b> |
| 13.1      | Influence functions for the two-by-two . . . . .          | 30        |
| 13.2      | Orthogonality and remainder of the DR-DiD score . . . . . | 30        |
| 13.3      | The Bacon decomposition . . . . .                         | 31        |
| 13.4      | Identification of $ATT(g,t)$ . . . . .                    | 31        |
| 13.5      | Multiplier bootstrap and simultaneous bands . . . . .     | 31        |
| 13.6      | References . . . . .                                      | 32        |

# 1 The problem

## 1.1 Same question, a new source of variation

Sections 03 and 04 identified treatment effects by comparing treated and untreated units that look alike. Selection on unobservables broke that strategy. Difference-in-differences (DiD) uses a second dimension: time. It compares the *change* in outcomes of treated units to the *change* among untreated units. Level differences between the groups cancel. Only the difference in trends must be assumed away.

That assumption is **parallel trends**: absent treatment, the treated group's average outcome would have moved like the untreated group's. It allows selection on anything fixed over time. It allows common shocks. It does not allow selection on future trends.

DiD is the right tool when three things hold.

- A group is exposed to a policy at a known date and another group is not.
- Both groups are observed before and after that date.
- The reason for exposure is not the group's outcome trajectory.

The data can be a panel (the same units over time) or repeated cross-sections (fresh draws from the same populations each period).

## 1.2 One experiment, the classic regression breaks three ways

In practice DiD is run as a two-way fixed effects (TWFE) regression,

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it},$$

with  $D_{it} = 1$  once unit  $i$  is treated. Section 2 shows that in the two-group, two-period case  $\beta$  is exactly the DiD of means. Applied work then extends the regression in three ways: adoption dates differ across units (staggered adoption), leads and lags of treatment are added to trace dynamics, and covariates are added linearly. Each extension can fail, and the failures are large.

**Design.** The staggered experiment follows Sant'Anna's lectures. There are 1000 units in 40 states. States adopt in one of four cohorts, 1986, 1992, 1998, or 2004. The panel runs from 1980 to 2003, so the 2004 cohort is untreated within the sample and serves as the comparison group. The effect grows with exposure and differs across cohorts:  $\tau_{it} = \mu_g(e + 1)$  at event time  $e = t - g$ , with  $\mu = (3, 2, 1)$  for the three treated cohorts. Untreated outcomes have unit effects, a common trend, and cohort-specific levels.

```
results <- run_staggered_experiment(n_sims = n_sims_staggered, n_units = 1000, seed = 1)
tab <- summarise_staggered_experiment(results)
pick <- function(estimator, e, column) tab[[column]][tab$estimator == estimator & tab$event_time == e]
```

**Estimators.** Five estimators of the event-study parameters, the average effect  $e$  periods after adoption for  $e = 0, \dots, 5$ .

| Estimator                           | What it does                                                                                             |
|-------------------------------------|----------------------------------------------------------------------------------------------------------|
| Static TWFE                         | one coefficient on $D_{it}$ , reported at every $e$                                                      |
| Dynamic TWFE, binned                | leads and lags of adoption, endpoints binned at $-5$ and $6+$                                            |
| Sun-Abraham                         | cohort-by-event-time coefficients, last cohort as comparison, aggregated                                 |
| Callaway-Sant'Anna, not yet treated | <code>att_gt()</code> cells with not-yet-treated comparisons, aggregated by <code>aggregate_att()</code> |
| Imputation (BJS)                    | <code>did_imputation()</code> : unit and time effects fitted on untreated cells, effects imputed         |

Table 2: Five estimators of the event-study parameters in the staggered design, 40 simulations.

| Event time | Estimator                           | Truth | Mean estimate | Bias  | SD   |
|------------|-------------------------------------|-------|---------------|-------|------|
| 0          | Static TWFE                         | 2.03  | 5.91          | 3.88  | 1.94 |
|            | Dynamic TWFE, binned                | 2.03  | 5.10          | 3.06  | 0.49 |
|            | Sun-Abraham                         | 2.03  | 2.04          | 0.01  | 0.19 |
|            | Callaway-Sant'Anna, not yet treated | 2.03  | 2.04          | 0.01  | 0.17 |
|            | Imputation (BJS)                    | 2.03  | 2.04          | 0.00  | 0.18 |
| 2          | Static TWFE                         | 6.09  | 5.91          | -0.18 | 1.94 |
|            | Dynamic TWFE, binned                | 6.09  | 6.42          | 0.33  | 0.34 |
|            | Sun-Abraham                         | 6.09  | 6.11          | 0.02  | 0.47 |
|            | Callaway-Sant'Anna, not yet treated | 6.09  | 6.11          | 0.02  | 0.47 |
|            | Imputation (BJS)                    | 6.09  | 6.11          | 0.02  | 0.47 |
| 4          | Static TWFE                         | 10.16 | 5.91          | -4.25 | 1.94 |
|            | Dynamic TWFE, binned                | 10.16 | 8.26          | -1.89 | 0.67 |
|            | Sun-Abraham                         | 10.16 | 10.14         | -0.01 | 0.79 |
|            | Callaway-Sant'Anna, not yet treated | 10.16 | 10.15         | 0.00  | 0.78 |
|            | Imputation (BJS)                    | 10.16 | 10.15         | -0.01 | 0.78 |

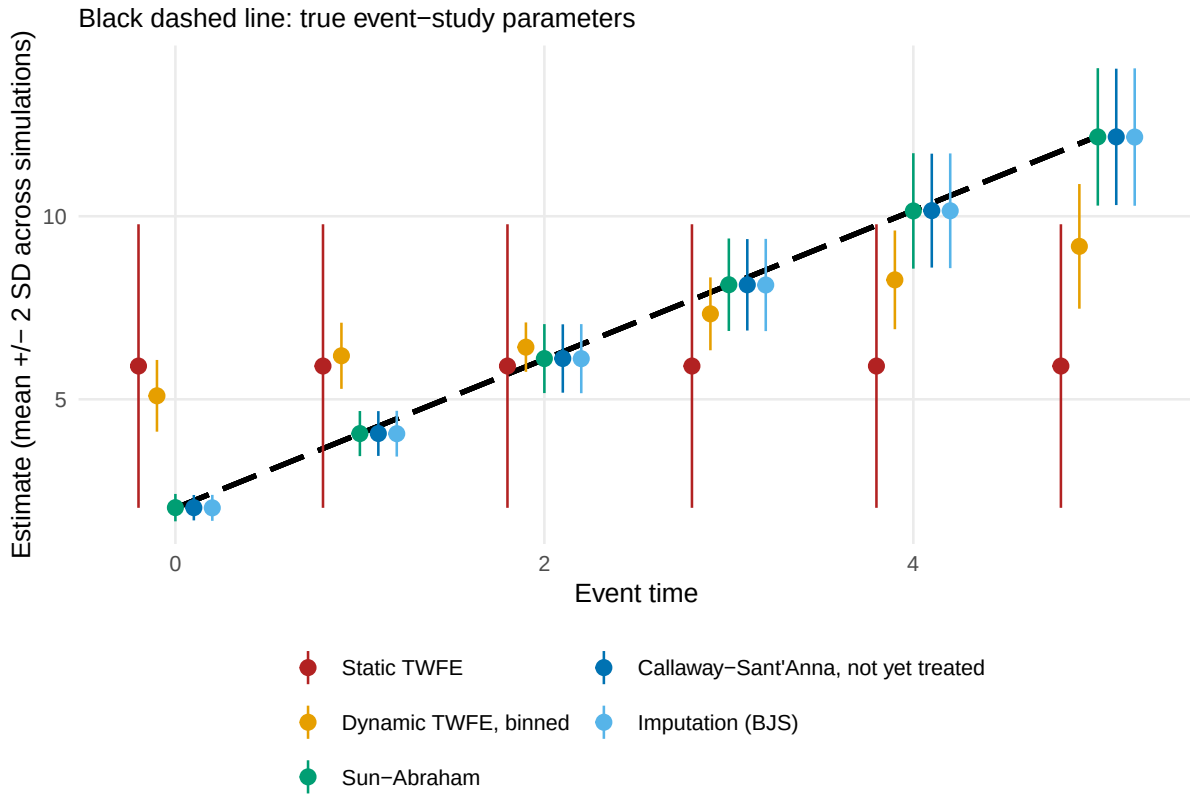


Figure 1: Mean estimate and a two-standard-deviation range across simulations. The two TWFE regressions miss the truth by a wide margin; the three estimators built from clean two-by-two comparisons track it.

The second failure needs only two periods. The design is the Kang-Schafer style example of Sant'Anna's lecture 5: four covariates drive both selection into treatment and the trend of untreated outcomes, and the true effect is zero.

```

cov_results <- run_covariate_experiment(n_sims = n_sims_covariates, n = 1000, seed = 1)
cov_tab <- summarise_covariate_experiment(cov_results, truth = 0)
pick_cov <- function(estimator, column) cov_tab[[column]][cov_tab$estimator == estimator]

```

Table 3: Two-period design with covariate-specific trends; true effect 0; 200 simulations.

| Estimator          | Bias   | SD   | RMSE  | Mean SE | 95% coverage |
|--------------------|--------|------|-------|---------|--------------|
| TWFE + X           | -16.54 | 2.27 | 16.69 | 2.23    | 0.00         |
| TWFE + post:X      | 0.01   | 0.09 | 0.09  | 0.10    | 0.96         |
| Outcome regression | 0.01   | 0.10 | 0.10  | 0.10    | 0.96         |
| IPW                | 0.07   | 1.54 | 1.53  | 1.42    | 0.92         |
| Doubly robust      | 0.01   | 0.10 | 0.10  | 0.11    | 0.96         |

### 1.3 Reading the results

- **Static TWFE** reports one number, 5.91, for every event time. It overstates the immediate effect by 3.88 and understates the effect four periods out by 4.25. Its spread across simulations is ten times that of the good estimators, because the weights it puts on cohorts and periods move with the realized cohort sizes. The regression averages many two-by-two comparisons, and some of them use already-treated units, whose effects are still growing, as controls (Section 5).
- **Dynamic TWFE** is biased at every event time, with bias 3.06 at  $e = 0$  and -1.89 at  $e = 4$ , and its pre-treatment coefficients are not zero either. Each coefficient is contaminated by effects from other event times and cohorts.
- **Sun-Abraham, Callaway-Sant’Anna, and imputation** are centred on the truth. They differ only in which comparisons they allow and which pre-periods they use (Section 6).
- **TWFE with covariates** in the two-period design has bias -16.54 and coverage 0.00. Interacting the covariates with the post indicator helps only if the trend is linear in  $X$ . Outcome regression, inverse probability weighting, and the doubly robust estimator all have coverage near 0.95 (Section 3).

None of these failures come from heterogeneity itself. They come from letting least squares choose the comparisons. The cure is the same throughout the lecture: write down the target, the assumption, and the comparison, and build the estimator from them.

### 1.4 Roadmap

Section 2 sets up potential outcomes, the assumptions, and the identification argument. Section 3 handles covariates. Section 4 studies dynamics with two groups. Sections 5 and 6 diagnose and fix staggered adoption. Section 7 asks how to assess parallel trends, Section 8 how to do inference. Section 9 covers continuous doses, treatments that switch off, triple differences, and nonlinear models. Section 10 covers synthetic DiD. Sections 11 and 12 close with replications, a checklist, and pointers.

## 2 Just enough framework

### 2.1 Potential outcomes indexed by adoption time

Units  $i = 1, \dots, n$  are observed in periods  $t = 1, \dots, T$ . Let  $G_i \in \{2, \dots, T\} \cup \{\infty\}$  be the period in which unit  $i$  is first treated, with  $G_i = \infty$  for units never treated in the sample. Treatment is absorbing: once treated, always treated (treatments that switch off are in Section 9). The potential outcome  $Y_{it}(g)$  is the outcome in period  $t$  if the unit were first treated in  $g$ . Observed outcomes are  $Y_{it} = \sum_g \mathbf{1}\{G_i = g\} Y_{it}(g)$ .

The building block of everything that follows is the **group-time average treatment effect**

$$ATT(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g], \quad t \geq g.$$

It is the effect at calendar time  $t$  of having started treatment at  $g$ , for the cohort that did start at  $g$ . The canonical two-by-two is the case  $T = 2$ ,  $G \in \{2, \infty\}$ , and  $ATT = ATT(2, 2)$ . Summaries across  $(g, t)$ , by exposure length, cohort, or calendar time, come in Section 6.

The target is an effect on the treated. Parallel trends will let us impute what treated units would have done without treatment; it says nothing about what untreated units would have done with it. The ATE needs a second assumption in the other direction, which is rarely credible.

## 2.2 Three assumptions

**Assumption 1 (no interference).**  $Y_{it} = \sum_g \mathbf{1}\{G_i = g\}Y_{it}(g)$ . One unit's treatment does not change another unit's outcome.

**Assumption 2 (no anticipation).**  $Y_{it}(g) = Y_{it}(\infty)$  for all  $t < g$ . Effects are zero before treatment starts.

**Assumption 3 (parallel trends).** For the two-by-two,

$$\mathbb{E}[Y_2(\infty) - Y_1(\infty) \mid G = 2] = \mathbb{E}[Y_2(\infty) - Y_1(\infty) \mid G = \infty].$$

With staggered adoption, for every treated cohort  $g$  and every  $t \geq g$ , either the *never-treated* version

$$\mathbb{E}[Y_t(\infty) - Y_{t-1}(\infty) \mid G = g] = \mathbb{E}[Y_t(\infty) - Y_{t-1}(\infty) \mid G = \infty],$$

or the *not-yet-treated* version, which replaces the right-hand side by the units still untreated at  $s \geq t$ :  $\mathbb{E}[\cdot \mid D_s = 0, G \neq g]$ . The *conditional* version adds “given  $X$ ” to both sides.

Three remarks.

- The two-by-two assumption restricts only the untreated potential outcome, and only its change. Levels may differ between groups by any amount; that is what “selection on fixed unobservables” means.
- The never-treated and not-yet-treated versions are different assumptions. The second uses more comparison units, so it is more precise, but it requires later cohorts to be valid controls for earlier ones. Both restrict post-treatment periods only. TWFE and the imputation estimator implicitly assume parallel trends in *all* periods, including before treatment (Section 6).
- With covariates, an overlap condition is needed:  $P(G = g \mid X) < 1$ , so every treated unit has comparable untreated units.

## 2.3 Identification of the 2x2 ATT: skeleton

Compare treated and untreated units after treatment:

$$\mathbb{E}[Y_2 \mid G = 2] - \mathbb{E}[Y_2 \mid G = \infty] = \underbrace{ATT}_{\text{target}} + \underbrace{\mathbb{E}[Y_2(\infty) \mid G = 2] - \mathbb{E}[Y_2(\infty) \mid G = \infty]}_{\text{selection bias}}.$$

The bias term contains the counterfactual  $\mathbb{E}[Y_2(\infty) \mid G = 2]$ . Three steps replace it by observables.

1. **Parallel trends** rewrites the counterfactual as the treated group's pre-period level plus the untreated group's change:  $\mathbb{E}[Y_2(\infty) \mid G = 2] = \mathbb{E}[Y_1(\infty) \mid G = 2] + \mathbb{E}[Y_2(\infty) - Y_1(\infty) \mid G = \infty]$ .
2. **No anticipation** turns  $Y_1(\infty)$  into  $Y_1(2)$  for the treated, which is the observed  $Y_1$ .
3. **No interference** turns every remaining potential outcome into an observed one.

Hence

$$ATT = (\mathbb{E}[Y_2 \mid G = 2] - \mathbb{E}[Y_1 \mid G = 2]) - (\mathbb{E}[Y_2 \mid G = \infty] - \mathbb{E}[Y_1 \mid G = \infty]).$$

The right-hand side is a difference of two before-after differences, hence the name. Appendix 13.4 writes the steps out.

**Estimation.** Replace expectations by sample means. In a panel, average the change  $\Delta Y_i = Y_{i2} - Y_{i1}$  within each group and take the difference. The same number is the coefficient on the interaction in the regression

$Y_{it} = \alpha + \gamma \mathbf{1}\{G_i = 2\} + \lambda \mathbf{1}\{t = 2\} + \beta \mathbf{1}\{G_i = 2\} \mathbf{1}\{t = 2\} + \varepsilon_{it}$ , the coefficient on the treated dummy in the regression of  $\Delta Y$  on it, and the coefficient on  $D_{it}$  in a TWFE regression with unit and time effects. All four agree to the last digit.

```
two <- sim_did_panel(n_units = 500, n_periods = 2, groups = 2, x_trend = 0, seed = 2)
two$treat <- as.integer(two$g == 2)
two$post <- as.integer(two$time == 2)
wide <- with(two, data.frame(id = id[time == 1], treat = treat[time == 1],
                             dy = y[time == 2] - y[time == 1]))
by_hand <- mean(wide$dy[wide$treat == 1]) - mean(wide$dy[wide$treat == 0])
c(by_hand = by_hand,
  interaction = coef(feols(y ~ treat * post, data = two))["treat:post"],
  first_difference = coef(feols(dy ~ treat, data = wide))["treat"],
  twfe = coef(feols(y ~ treat:post | id + time, data = two))["treat:post"],
  att_gt = att_gt(two, id = "id", time = "time", group = "g", y = "y", n_boot = 0)$att_gt$att)
#>      by_hand      interaction first_difference      twfe
#>      3.282635      3.282635      3.282635      3.282635
#>      att_gt
#>      3.282635
```

The regression is a device for computing means. It does not change the target or the assumption. Once the design has more than two groups or two periods, that convenience disappears: the regression starts choosing comparisons on its own (Section 5).

## 2.4 From two periods to long differences

With several post-treatment periods, apply the argument period by period. Parallel trends from  $t - 1$  to  $t$  for every  $t \geq g$  chains into

$$\mathbb{E}[Y_t(\infty) \mid G = g] = \mathbb{E}[Y_{g-1} \mid G = g] + \mathbb{E}[Y_t - Y_{g-1} \mid G = \infty],$$

so

$$ATT(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid G = \infty].$$

The comparison is a **long difference** from the last pre-treatment period  $g - 1$  to  $t$ . Intermediate periods drop out; they are used only through the assumption. Before treatment, the same assumptions imply that the short difference from  $t - 1$  to  $t$  has zero DiD, which is the placebo check of Section 7.

```
one <- sim_did_panel(n_units = 400, n_periods = 6, groups = 4, never_share = 0.5, seed = 3)
long_diff <- one %>%
  group_by(id, g) %>%
  summarise(ld = y[time == 6] - y[time == 3], .groups = "drop") %>%
  summarise(att = mean(ld[g == 4]) - mean(ld[g == 0])) %>% pull(att)
cell <- att_gt(one, id = "id", time = "time", group = "g", y = "y", n_boot = 0)$att_gt
c(long_difference_by_hand = long_diff, att_gt_4_6 = cell$att[cell$group == 4 & cell$time == 6])
#> long_difference_by_hand      att_gt_4_6
#>      8.962091      8.962091
```

## 2.5 What parallel trends really assumes

Parallel trends is a statement about a specific transformation of the outcome. If it holds for  $Y$  it generally fails for  $\log Y$ , and the two DiDs answer different questions with different assumptions. Roth and Sant'Anna (2023) show that parallel trends holds for every monotone transformation only when it holds for the whole distribution of untreated outcomes, and that this happens only when the population is a mixture of two types: units for which treatment is as good as random, and units whose untreated outcome distribution does not change over time.

A two-line example. Treated units have higher outcome levels. Nothing happens over time except a common additive shift, so parallel trends holds in levels and the true effect is zero.

```
set.seed(4)
n <- 4000
treat <- rep(c(0, 1), each = n / 2)
y1 <- exp(rnorm(n, mean = ifelse(treat == 1, 2, 1), sd = 0.5))
```

```

y2 <- y1 + 3
did <- function(v1, v2) mean((v2 - v1)[treat == 1]) - mean((v2 - v1)[treat == 0])
c(levels = did(y1, y2), logs = did(log(y1), log(y2)))
#>      levels      logs
#> 0.0000000 -0.4106026

```

The DiD in logs is far from zero. A common shift of 3 units is a smaller proportional change for the group with higher levels, so the log trends are not parallel. The choice of scale is part of the identifying assumption and should be argued from the economics of the outcome, not from fit. Section 7 shows a test that can reject the joint hypothesis that parallel trends holds on every scale.

## 2.6 Inference from influence functions

Write the panel two-by-two estimator as  $\hat{\theta} = \bar{\Delta Y}_{G=2} - \bar{\Delta Y}_{G=\infty}$ . A first-order expansion gives

$$\sqrt{n}(\hat{\theta} - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i + o_p(1), \quad \xi_i = \frac{\mathbf{1}\{G_i = 2\}}{P(G=2)} (\Delta Y_i - \mathbb{E}[\Delta Y | G=2]) - \frac{\mathbf{1}\{G_i = \infty\}}{P(G=\infty)} (\Delta Y_i - \mathbb{E}[\Delta Y | G=\infty]).$$

$\xi_i$  is the **influence function**: unit  $i$ 's contribution to the estimation error. It has mean zero, so the central limit theorem gives  $\sqrt{n}(\hat{\theta} - \theta) \rightarrow N(0, \mathbb{E}[\xi^2])$ , and the standard error is  $\sqrt{\sum_i \hat{\xi}_i^2 / n}$ . Three things follow.

- **Clustering** at a level above the unit sums  $\xi_i$  within clusters before squaring. The unit is the smallest admissible level, because the two periods of a unit are one draw.
- **Many parameters** share one machinery. Every  $ATT(g, t)$  has an influence function, and any weighted average of cells has the corresponding weighted average of influence functions. Covariances come free.
- **The multiplier bootstrap** draws i.i.d. weights  $V_i$  with mean zero and variance one and forms  $\hat{\theta}^* = \hat{\theta} + n^{-1} \sum_i V_i \hat{\xi}_i$ . No re-estimation is needed. The distribution of  $\max_k |\hat{\theta}_k^* - \hat{\theta}_k| / \hat{se}_k$  across draws gives the critical value of a **simultaneous confidence band** over all cells.

`att_gt()` returns the influence functions and uses them for all of this. On the two-period data, the influence-function standard error equals the heteroskedasticity-robust standard error of the first-difference regression up to the finite-sample factor, and the bootstrap agrees with both.

```

fit0 <- att_gt(two, id = "id", time = "time", group = "g", y = "y", n_boot = 0)
fitb <- att_gt(two, id = "id", time = "time", group = "g", y = "y", n_boot = 999, seed = 1)
xi <- fit0$inffunc[, 1]
c(influence_function = sqrt(sum(xi^2)) / nrow(wide),
  att_gt_analytic = fit0$att_gt$std.error,
  fixest_hc1 = se(feols(dy ~ treat, data = wide, vcov = "hetero"))[["treat"]],
  multiplier_bootstrap = fitb$att_gt$std.error)
#> influence_function att_gt_analytic fixest_hc1
#> 0.1251485 0.1251485 0.1253996
#> multiplier_bootstrap
#> 0.1216301

```

Appendix 13.1 derives  $\xi_i$  for the panel and the repeated cross-section cases and states the bootstrap result.

## 3 Covariates: three faces of conditional parallel trends

### 3.1 Why covariates

Parallel trends is often more plausible within groups of similar units than overall. Employment in northern and southern states, earnings of young and old workers, sales of large and small retailers all trend differently. If treated units are richer in the fast-trending type, the unconditional comparison confounds the effect with composition. The fix is to assume parallel trends **conditional on** pre-treatment covariates  $X$ :

$$\mathbb{E}[Y_2(\infty) - Y_1(\infty) | G = 2, X] = \mathbb{E}[Y_2(\infty) - Y_1(\infty) | G = \infty, X].$$



Two rules. Covariates must be measured before treatment, or be unaffected by it; controlling for a post-treatment variable blocks part of the effect. And overlap is needed:  $P(G = 2 | X) < 1$ , so every treated unit has untreated units with similar  $X$ .

A classic case is the **Ashenfelter dip**: people enter training after a bad year, so their pre-period earnings are transitorily low and mean-revert afterwards. Conditioning on a longer earnings history, or moving the base period back, is the remedy; a two-period DiD with the dip year as the base is not.

### 3.2 The tempting regression and why it fails

The natural move is to add  $X$  to the regression,

$$Y_{it} = \alpha + \gamma \mathbf{1}\{G_i = 2\} + \lambda \mathbf{1}\{t = 2\} + \beta \mathbf{1}\{G_i = 2\} \mathbf{1}\{t = 2\} + X_i' \delta + \varepsilon_{it}.$$

Read off its four conditional means. The untreated change is  $\lambda$  for every value of  $X$ : the regression assumes there are no covariate-specific trends, which is exactly what the covariates were supposed to allow. The treated change is  $\lambda + \beta$  for every  $X$ : the effect is assumed homogeneous across covariate values. When either restriction fails,  $\beta$  is not the ATT. The Section 1 table showed a bias of -16.54 with zero coverage.

Interacting  $X$  with the post indicator allows covariate-specific trends that are linear in  $X$ . That worked in Section 1 because the design was linear. It is a modelling assumption, not a fix.

```
kd <- simulate_covariate_dgp(n = 1000, seed = 5)
kd$treat <- as.integer(kd$g == 2); kd$post <- as.integer(kd$time == 2)
c(twfe_plus_x = coef(feols(y ~ treat:post + x1 + x2 + x3 + x4 | id + time, data = kd))["treat:post"],
  twfe_post_x = coef(feols(y ~ treat:post + post:(x1 + x2 + x3 + x4) | id + time, data = kd))["treat:post"],
  dr = att_gt(kd, id = "id", time = "time", group = "g", y = "y",
    x = c("x1", "x2", "x3", "x4"), n_boot = 0)$att_gt$att)
#> twfe_plus_x twfe_post_x dr
#> -16.98849256 0.11203544 0.06496616
```

### 3.3 Outcome regression, IPW, doubly robust

Separate identification from estimation. Under conditional parallel trends, with  $D = \mathbf{1}\{G = 2\}$ ,  $\Delta Y = Y_2 - Y_1$ ,  $p(X) = P(D = 1 | X)$  and  $m_\Delta(X) = \mathbb{E}[\Delta Y | D = 0, X]$ , the ATT has three equivalent expressions.

| Face                                              | Estimand                                                                                                                                                       | What must be modelled          |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Outcome regression (Heckman, Ichimura, Todd 1997) | $\mathbb{E}[\Delta Y   D = 1] - \mathbb{E}[m_\Delta(X)   D = 1]$                                                                                               | the untreated change given $X$ |
| Inverse probability weighting (Abadie 2005)       | $\mathbb{E}[w_1 \Delta Y] - \mathbb{E}[w_0 \Delta Y]$ , $w_1 = \frac{D}{\mathbb{E}[D]}$ ,<br>$w_0 = \frac{p(X)(1-D)/(1-p(X))}{\mathbb{E}[p(X)(1-D)/(1-p(X))]}$ | the propensity score           |
| Doubly robust (Sant'Anna and Zhao 2020)           | $\mathbb{E}[(w_1 - w_0)(\Delta Y - m_\Delta(X))]$                                                                                                              | either one correctly           |

The doubly robust estimand is the IPW estimand applied to the residual  $\Delta Y - m_\Delta(X)$ . If the outcome model is right, the residual has conditional mean zero among the untreated, so any weights work. If the propensity score is right, the weights balance  $X$ , so any outcome model works. When both are right it reaches the semiparametric efficiency bound. `att_gt(method = )` implements the three with a logit and a linear regression as working models, reproducing the DRDID package.

```
sapply(c("reg", "ipw", "dr"), function(m)
  att_gt(kd, id = "id", time = "time", group = "g", y = "y",
    x = c("x1", "x2", "x3", "x4"), method = m, n_boot = 0)$att_gt[, c("att", "std.error")])
#>      reg      ipw      dr
#> att      0.1061084 1.907314 0.06496616
#> std.error 0.09984184 1.205576 0.1019301
```

The guarantee is about *specification*, not about *fit*. When the true drivers are nonlinear transformations of  $X$  and the analyst enters  $X$  linearly in both models, nothing is guaranteed.

```

mis_results <- run_covariate_experiment(n_sims = 100, n = 1000, seed = 2, misspecify = TRUE)
mis_tab <- summarise_covariate_experiment(mis_results, truth = 0)
pick_mis <- function(estimator, column) mis_tab[[column]][mis_tab$estimator == estimator]

```

Table 5: Both working models misspecified (the true drivers are nonlinear in X); true effect 0; 100 simulations.

| Estimator          | Bias   | SD   | RMSE  | Mean SE | 95% coverage |
|--------------------|--------|------|-------|---------|--------------|
| TWFE + X           | -12.19 | 2.10 | 12.37 | 2.49    | 0.00         |
| Outcome regression | -0.01  | 1.61 | 1.60  | 1.56    | 0.94         |
| IPW                | -1.23  | 1.92 | 2.27  | 1.74    | 0.89         |
| Doubly robust      | -4.09  | 2.09 | 4.59  | 1.88    | 0.37         |

In this design the linear outcome model happens to extrapolate well, so outcome regression is nearly unbiased (bias -0.01). The misspecified logit produces extreme weights, IPW is biased (-1.23), and the doubly robust estimator, which applies those weights to the residuals, is the most biased of the three (-4.09, coverage 0.37). Double robustness insures against one wrong model. With two wrong models it can be worse than either alone. The way out is flexible estimation of both nuisances, which is the next subsection.

### 3.4 DR-DiD is the ATT score of Section 04 on the differenced outcome

Compare the doubly robust estimand with the ATT score of Section 04. With  $\Delta Y$  in place of  $Y$  and  $m_\Delta$  in place of  $\mu_0$ , they are the same object: the augmented inverse probability weighting score for the average effect on the treated. The only difference is the normalization of the control weights, by their own mean here and by  $\mathbb{E}[D]$  in Section 04, which is immaterial in large samples. Everything Section 04 established therefore carries over: the score is Neyman orthogonal, its remainder is the product of the two nuisance errors, and machine learners with cross-fitting may estimate the nuisances.

```

wide_kd <- with(kd, data.frame(id = id[time == 1], d = treat[time == 1],
                             dy = y[time == 2] - y[time == 1],
                             x1 = x1[time == 1], x2 = x2[time == 1], x3 = x3[time == 1], x4 = x4[time == 1]))
p_hat <- fitted(glm(d ~ x1 + x2 + x3 + x4, data = wide_kd, family = binomial))
m_hat <- predict(lm(dy ~ x1 + x2 + x3 + x4, data = wide_kd[wide_kd$d == 0, ], newdata = wide_kd))
c(att_gt_dr = att_gt(kd, id = "id", time = "time", group = "g", y = "y",
                    x = c("x1", "x2", "x3", "x4"), n_boot = 0)$att_gt$att,
  est_dml_att = est_dml(wide_kd, y = "dy", d = "d", model = "irm", estimand = "ATT",
                       p_hat = p_hat, mu0_hat = m_hat, mu1_hat = m_hat)$estimate)
#> att_gt_dr est_dml_att
#> 0.06496616 0.06346630

```

### 3.5 Machine-learning nuisances and cross-fitting

Supplying mlr3 learners to att\_gt() switches each cell to cross-fitted nuisances: the propensity score and the untreated change are predicted out of fold, and the doubly robust score is evaluated on the held-out units. This is the recipe of Section 04 inside each  $(g, t)$  cell. It needs learners that converge fast enough, roughly  $n^{-1/4}$  for the product of the two errors, and it pays off when the functional forms are unknown, as in the misspecified design.

```

library(mlr3); library(mlr3learners)
mis <- simulate_covariate_dgp(n = 2000, seed = 6, misspecify = TRUE)
rbind(
  parametric_dr = att_gt(mis, id = "id", time = "time", group = "g", y = "y",
                        x = c("x1", "x2", "x3", "x4"), n_boot = 0)$att_gt[, c("att", "std.error")],
  forest_dr = att_gt(mis, id = "id", time = "time", group = "g", y = "y", x = c("x1", "x2", "x3", "x4"),
                    learner_p = lrn("classif.ranger", predict_type = "prob", num.trees = 300),
                    learner_or = lrn("regr.ranger", num.trees = 300),
                    folds = 5, seed = 1, n_boot = 0)$att_gt[, c("att", "std.error")]
)
#> att std.error

```

```
#> parametric_dr -8.1856572 3.8678025
#> forest_dr      0.5584006 0.5695338
```

### 3.6 Repeated cross-sections

With repeated cross-sections the same units are not observed twice, so  $\Delta Y$  does not exist. The estimators use four conditional means, one per group and period, and need one more assumption: the composition of each group does not change between periods (stationarity). Two doubly robust versions exist. The first models only the untreated outcomes in each period; the second also models the treated outcomes, which makes it efficient (Sant’Anna and Zhao 2020). `att_gt(sampling = "rcs")` uses the second. For a given sample size, repeated cross-sections are always less precise than a panel, and more so when the pre and post samples are unbalanced.

```
set.seed(7)
rcs <- kd[sample(nrow(kd), 1500), ]
rcs$row <- seq_len(nrow(rcs))
rbind(
  panel = att_gt(kd, id = "id", time = "time", group = "g", y = "y",
    x = c("x1", "x2", "x3", "x4"), n_boot = 0)$att_gt[, c("att", "std.error", "n_treated", "n_control")],
  repeated_cross_section = att_gt(rcs, id = "row", time = "time", group = "g", y = "y",
    x = c("x1", "x2", "x3", "x4"), sampling = "rcs", n_boot = 0)$att_gt[, c("att", "std.error", "n_treated", "n_control")]
)
#>               att std.error n_treated n_control
#> panel          0.06496616 0.1019301      498      502
#> repeated_cross_section 0.16069843 0.1775012      751      749
```

## 4 Dynamics with two groups

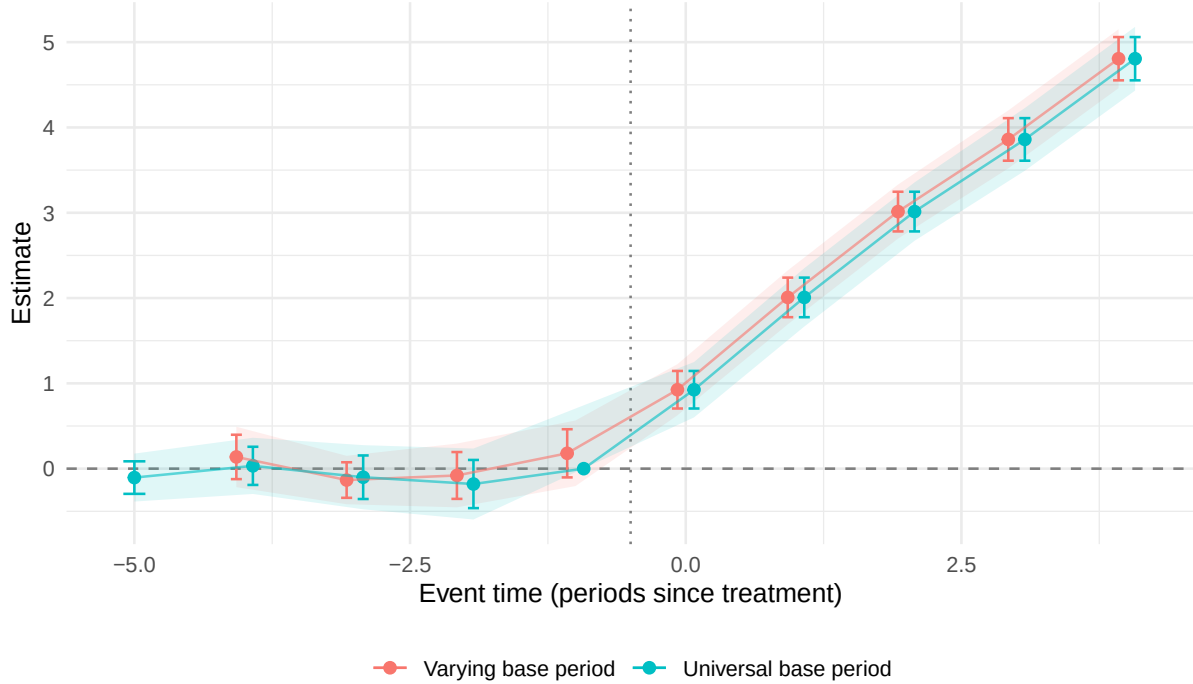
### 4.1 ATT( $g, t$ ) as an event study

One treated cohort  $g$ , one comparison group, many periods. Section 2.4 identified  $ATT(g, t)$  for  $t \geq g$  by the long difference from  $g - 1$ . Indexing by event time  $e = t - g$  turns the cells into an **event study**: the effect  $e$  periods after adoption. Cells with  $e < 0$  are placebos. Under no anticipation and parallel trends they equal zero, and they are what a pre-trends plot shows.

Two conventions exist for the pre-treatment cells.

- **Varying base period** (default): the cell at  $e < 0$  compares  $t - 1$  to  $t$ , a short difference. It shows period-by-period deviations from parallel trends.
- **Universal base period**: every cell is compared to  $g - 1$ . Post cells are unchanged. Pre cells show the cumulative deviation between  $t$  and  $g - 1$ , so the plot is anchored at  $e = -1$  and reads like a standard event-study figure. Use it for sensitivity analysis (Section 7).

```
esd <- sim_did_panel(n_units = 500, n_periods = 10, groups = 6, never_share = 0.5, mu = 1, seed = 8)
es_var <- att_gt(esd, id = "id", time = "time", group = "g", y = "y", control_group = "never",
  base_period = "varying", n_boot = 199, seed = 1)
es_uni <- att_gt(esd, id = "id", time = "time", group = "g", y = "y", control_group = "never",
  base_period = "universal", n_boot = 199, seed = 1)
plot_event_study(event_study_frame("Varying base period" = es_var, "Universal base period" = es_uni))
```



The post-treatment cells coincide. The pre-treatment cells differ only in what they cumulate; both are centred at zero here because parallel trends holds in the design.

## 4.2 Anticipation

If units respond before the official start, because the policy was announced or phased in, no anticipation fails for the periods just before  $g$ . The base period  $g - 1$  is then already “treated”, and every post effect is measured against a contaminated baseline. The remedy is **limited anticipation**: assume no effects before  $g - k$  for a known  $k$ , strengthen parallel trends to hold from  $g - k - 1$  on, and use  $g - k - 1$  as the base period. `att_gt(anticipation = k)` does exactly that.

```
ant <- sim_did_panel(n_units = 600, n_periods = 8, groups = 5, never_share = 0.5, mu = 2,
                    anticipation = 1, seed = 9)
truth_ant <- attr(ant, "att_gt")
cmp <- lapply(0:1, function(k) {
  f <- att_gt(ant, id = "id", time = "time", group = "g", y = "y", control_group = "never",
             anticipation = k, n_boot = 0)$att_gt
  data.frame(anticipation = k, event_time = f$event_time, att = round(f$att, 2))
})
tab_ant <- merge(cmp[[1]], cmp[[2]], by = "event_time", suffixes = c("_k0", "_k1"))
tab_ant$truth <- ifelse(tab_ant$event_time >= 0, truth_ant$att[match(tab_ant$event_time + 5, truth_ant$time)],
                      ifelse(tab_ant$event_time == -1, 0.5 * truth_ant$att[1], 0))
tab_ant[tab_ant$event_time >= -2 & tab_ant$event_time <= 2, c("event_time", "att_k0", "att_k1", "truth")]
#>   event_time att_k0 att_k1 truth
#> 2         -2 -0.06 -0.06    0
#> 3         -1  1.05  1.05    1
#> 4          0  0.99  2.04    2
#> 5          1  3.05  4.10    4
#> 6          2  5.02  6.07    6
```

In the design, half of the first-period effect appears one period early. With `anticipation = 0` every post-treatment effect is understated by that amount, and the cell at  $e = -1$  looks like a pre-trend. With `anticipation = 1` the base moves to  $g - 2$ , the post effects are right, and the cell at  $e = -1$  correctly measures the anticipation effect. The lesson is to date treatment at the announcement, or to set  $k$  from institutional knowledge, not from the plot.

### 4.3 TWFE event-study regressions

The regression version adds leads and lags of adoption,

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \gamma_k \mathbf{1}\{t - G_i = k\} + \varepsilon_{it},$$

with the period before adoption omitted. With one cohort, a never-treated comparison group, and **all** leads and lags included, the model is saturated and  $\hat{\gamma}_k$  equals the universal-base  $ATT(g, g+k)$  of `att_gt()` exactly. Binning the endpoints, for example one dummy for  $k \leq -4$  and one for  $k \geq 3$ , is an extra restriction: it assumes the effect is constant inside the bin, and the other coefficients change.

```
esd$rel <- ifelse(esd$g > 0, esd$time - esd$g, -1000)
tw_all <- feols(y ~ i(rel, ref = c(-1, -1000)) | id + time, data = esd, cluster = ~id)
esd$rel_bin <- ifelse(esd$g == 0, -1000, pmin(pmax(esd$rel, -3), 2))
tw_bin <- feols(y ~ i(rel_bin, ref = c(-1, -1000)) | id + time, data = esd, cluster = ~id)
fr <- event_study_frame("att_gt" = es_uni, "TWFE_all" = tw_all, "TWFE_binned" = tw_bin)
fr %>% filter(event_time %in% c(-3, -2, 0, 1, 2)) %>%
  select(method, event_time, estimate) %>%
  tidyr::pivot_wider(names_from = method, values_from = estimate) %>%
  as.data.frame()
#>   event_time   att_gt TWFE_all TWFE_binned
#> 1          -3 -0.1004025 -0.1004025 -0.05732787
#> 2           -2 -0.1800596 -0.1800596 -0.18005960
#> 3            0  0.9253220  0.9253220  0.92532198
#> 4            1  2.0076674  2.0076674  2.00766740
#> 5            2  3.0137673  3.0137673  3.89314473
```

The saturated regression and `att_gt()` agree to the last digit. The binned regression does not, and the difference is not a bug but a restriction. Report all event times and let the reader choose what to look at.

### 4.4 Simultaneous confidence bands

An event study is many parameters. Pointwise 95 percent intervals cover each cell 95 percent of the time, but the probability that *all* cells are covered at once is much lower, and the eye reads the curve as a whole. A **simultaneous band** fixes the joint coverage. `att_gt()` computes it from the multiplier bootstrap: draw the joint distribution of the studentized cells, take the maximum absolute deviation across cells, and use its 95th percentile as the critical value. The band is wider than the pointwise interval by that factor.

```
c(pointwise = qnorm(0.975), simultaneous = es_uni$crit_val)
#>   pointwise simultaneous
#> 1.959964  2.887939
es_uni$pretest
#> $statistic
#> [1] 3.576562
#>
#> $df
#> [1] 4
#>
#> $p.value
#> [1] 0.4663329
```

The pre-test is a joint Wald test that all pre-treatment cells are zero, computed from the covariance of their influence functions. It answers a narrower question than it seems to; Section 7 returns to it.

## 5 Staggered adoption: what goes wrong

### 5.1 Static TWFE as a weighted average

With several adoption dates, the static regression  $Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}$  still computes a difference-in-differences, but a weighted average of every two-by-two comparison the data allow (Goodman-Bacon 2021):

$$\hat{\beta} = \sum_k s_{kU} \hat{\beta}_{kU} + \sum_{k < l} (s_{kl}^k \hat{\beta}_{kl}^k + s_{kl}^l \hat{\beta}_{kl}^l).$$

The three types are: cohort  $k$  against never-treated units; the earlier cohort  $k$  against the later cohort  $l$  before  $l$  adopts, where  $l$  is a clean control; and the later cohort  $l$  against the earlier cohort  $k$  after  $k$  has adopted. In the third type the control group is already treated. If its effect is still changing, its trend is not the untreated trend, and the comparison is biased. The weights are positive and sum to one, but they depend on group sizes and on how central each adoption date is in the panel, not on anything a researcher would choose.

```
sd1 <- simulate_staggered_dgp(n_units = 1000, seed = 11)
bd <- bacon_decomp(sd1, id = "id", time = "year", y = "y", d = "treated")
bd$by_type
#>           type      weight estimate
#> 1 earlier vs later treated 0.2010874 11.48134
#> 2 later vs earlier treated 0.2072902 -15.47035
#> 3 treated vs never treated 0.5916223 14.77469
c(twfe = bd$twfe, weighted_sum = bd$check)
#>           twfe weighted_sum
#>      7.842941      7.842941
```

The later-versus-earlier comparisons carry a substantial weight and a negative estimate, although every true effect is positive. That is the mechanism behind the Section 1 result.

## 5.2 Negative weights

de Chaisemartin and D'Haultfoeulle (2020) give a second reading. Under parallel trends, the same coefficient is a weighted sum of the treatment effects of the treated unit-periods,  $\beta = \sum_{(i,t): D_{it}=1} w_{it} \tau_{it}$ , with weights that sum to one but can be negative. The weight of a cell is the residual of  $D_{it}$  on unit and period effects, divided by the sum of those residuals. Cells treated for a long time in periods when most units are treated get negative weights, because their treatment status is *below* what the two fixed effects predict.

```
tw <- twfe_weights(sd1, id = "id", time = "year", d = "treated", y = "y")
c(share_negative = tw$share_negative, sum_negative = tw$sum_negative)
#> share_negative sum_negative
#>      0.14174894      -0.06025356
tw$by_cell %>% group_by(g) %>%
  summarise(first_year = min(time), last_year = max(time), min_cell_weight = min(weight),
            n_negative = sum(weight < 0), cohort_weight = sum(weight), .groups = "drop") %>%
  as.data.frame()
#>      g first_year last_year min_cell_weight n_negative cohort_weight
#> 1 1986      1986      2003      -0.01004226          6      0.2576247
#> 2 1992      1992      2003       0.02713719          0      0.5086285
#> 3 1998      1998      2003       0.03895781          0      0.2337469
```

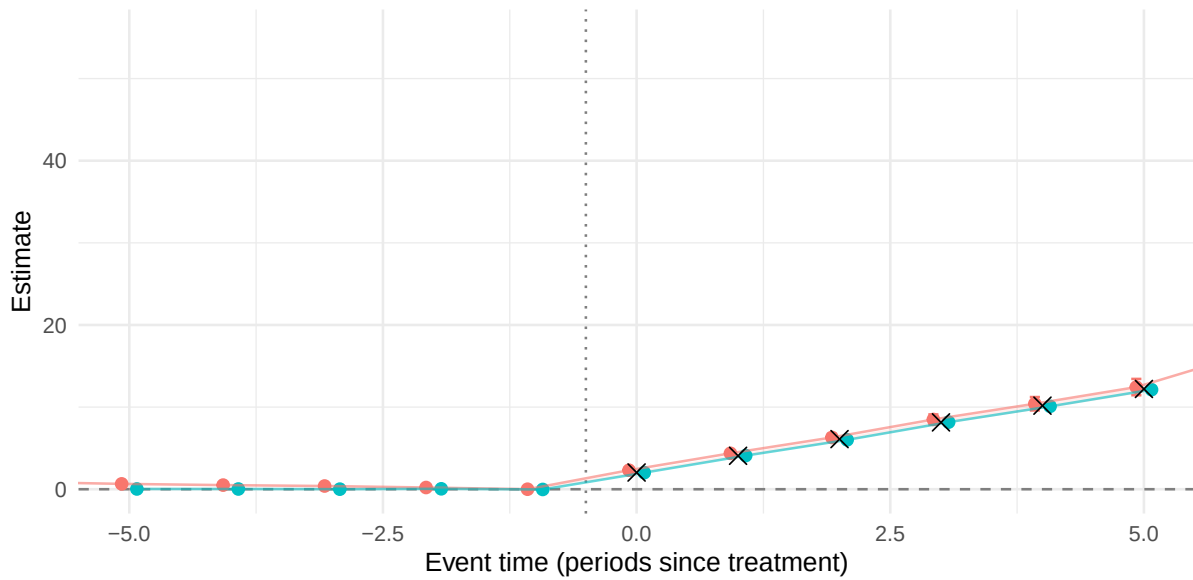
A negative weight on a positive effect pulls the estimate down. When effects grow with exposure, the negative weights fall on the largest effects, and the coefficient can even change sign.

## 5.3 Dynamic TWFE and contamination

Adding leads and lags does not help. Sun and Abraham (2021) show that with heterogeneous effects across cohorts, the coefficient on lag  $k$  is a weighted average of  $ATT(g, g + k)$  across cohorts, with weights that need not be convex, *plus* contamination terms from every other lag and lead. Two consequences matter in practice. Pre-treatment coefficients can be nonzero even when no anticipation and parallel trends hold, so a pre-trends plot from this regression can reject a valid design. And binning the endpoints, which imposes constant effects far from adoption, moves the contamination into the window of interest.

```
sd1$rel <- ifelse(sd1$g > 0, sd1$year - sd1$g, -1000)
tw_dyn <- feols(y ~ i(rel, ref = c(-1, -1000)) | id + year, data = sd1, cluster = ~state)
cs1 <- att_gt(sd1, id = "id", time = "year", group = "g", y = "y", control_group = "notyet", n_boot = 0)
cs1_dyn <- aggregate_att(cs1, type = "dynamic", min_e = -5, max_e = 5, n_boot = 0)
truth_sd1 <- attr(sd1, "truth")
plot_event_study(event_study_frame("TWFE, all leads and lags" = tw_dyn, "Callaway-Sant'Anna" = cs1_dyn)) +
  geom_point(data = truth_sd1, aes(x = event_time, y = att), inherit.aes = FALSE, shape = 4, size = 3) +
  coord_cartesian(xlim = c(-5, 5)) +
  labs(subtitle = "Crosses: true event-study parameters; no anticipation in the design")
```

Crosses: true event-study parameters; no anticipation in the design



— TWFE, all leads and lags — Callaway-Sant'Anna

```
event_study_frame("TWFE" = tw_dyn, "CS" = cs1_dyn) %>%
  filter(event_time %in% c(-4, -2, 0, 2, 4)) %>%
  select(method, event_time, estimate) %>%
  tidyr::pivot_wider(names_from = method, values_from = estimate) %>%
  left_join(truth_sd1, by = "event_time") %>% as.data.frame()

#>   event_time TWFE      CS      att
#> 1         -4  0.5013917 0.02725957    NA
#> 2          -2  0.2082017 0.05018534    NA
#> 3           0  2.3116429 2.01589419 2.034433
#> 4           2  6.3030461 6.00965189 6.103300
#> 5           4 10.4073786 10.08898161 10.172166
```

The regression shows a pre-trend that does not exist and misstates the post-treatment effects. The cell-based estimator shows neither.

## 5.4 What is not the problem

None of this is caused by heterogeneity as such. If effects were the same for every cohort and constant over time, the forbidden comparisons would be harmless and static TWFE would be fine.

```
sd0 <- simulate_staggered_dgp(n_units = 1000, mu = c(2, 2, 2, 2), seed = 12)
sd0$y <- sd0$y - sd0$tau + 2 * sd0$treated # constant effect of 2
c(twfe = coef(feols(y ~ treated | id + year, data = sd0))["treated"]),
  cs = aggregate_att(att_gt(sd0, id = "id", time = "year", group = "g", y = "y", n_boot = 0),
    type = "simple", n_boot = 0)$overall$estimate)

#>   twfe      cs
#> 1.977243 1.933191
```

The problem is that least squares chooses the comparisons and the weights. Heterogeneous dynamics are the normal case in economics and marketing, so the estimator must be built to survive them. That is the next section.

## 6 Staggered adoption: reliable estimators

### 6.1 Identification, aggregation, inference as three steps

Every reliable estimator in this section follows the same order, which is the reverse of “run the regression and interpret the coefficient”.

1. **Identify** each  $ATT(g, t)$  from a stated parallel trends assumption and a stated comparison group (Section 2.4).
2. **Aggregate** the cells with weights chosen for the question, by exposure length, cohort, or calendar time.
3. **Infer** from the influence functions of the cells, so that the aggregate inherits valid standard errors and simultaneous bands.

Least squares performs all three steps at once with weights nobody chose. Separating them is the whole content of Callaway and Sant’Anna (2021).

### 6.2 The building block $ATT(g, t)$

`att_gt()` estimates every cell. The comparison group is a choice with consequences.

- **Never treated:** one fixed comparison group. Parallel trends is assumed between each cohort and the never-treated units only.
- **Not yet treated:** every unit still untreated at the later of  $t$  and the base period. More units, hence more precision, at the price of assuming parallel trends among all cohorts as well.

```
cs_never <- att_gt(sdl, id = "id", time = "year", group = "g", y = "y", control_group = "never",
  n_boot = 199, seed = 1)
cs_notyet <- att_gt(sdl, id = "id", time = "year", group = "g", y = "y", control_group = "notyet",
  n_boot = 199, seed = 1)
tidy(cs_never) %>% filter(group == 1992, event_time %in% 0:3) %>%
  mutate(se_notyet = tidy(cs_notyet)$std.error[match(term, tidy(cs_notyet)$term)]) %>%
  select(group, time, event_time, estimate, se_never = std.error, se_notyet)
#>   group time event_time estimate se_never se_notyet
#> 1  1992 1992          0 1.836580 0.1304550 0.1178785
#> 2  1992 1993          1 4.061156 0.1363729 0.1223339
#> 3  1992 1994          2 5.844356 0.1293339 0.1238733
#> 4  1992 1995          3 7.947868 0.1354121 0.1258641

tidy(cs_notyet) %>%
  ggplot(aes(x = time, y = estimate, colour = factor(post))) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_ribbon(aes(ymin = band.low, ymax = band.high, group = 1), fill = "grey70", alpha = 0.3, colour = NA) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high), size = 0.3) +
  facet_wrap(~ group, labeller = label_both) +
  scale_colour_manual(values = c("0" = "grey40", "1" = "#0072B2"), labels = c("pre-treatment", "post-treatment")) +
  labs(x = "Year", y = "ATT(g, t)", colour = NULL) + did_theme()
```



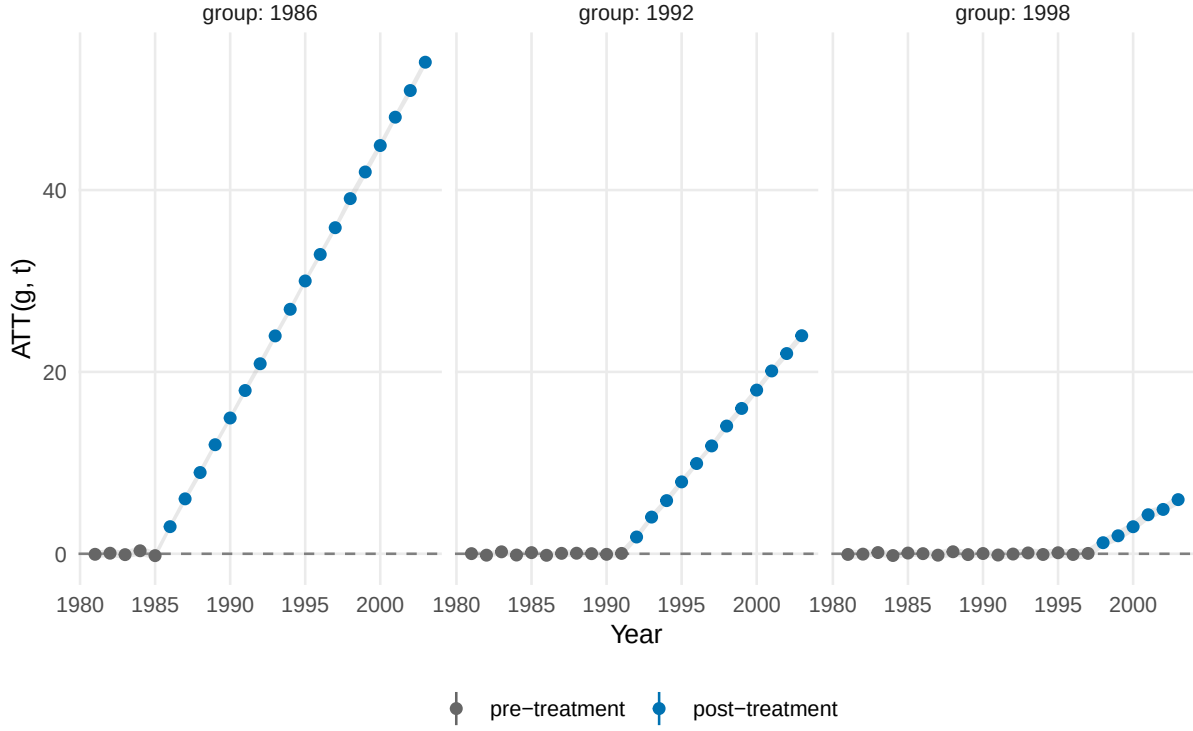


Figure 2: Group-time effects with not-yet-treated comparisons; error bars are pointwise intervals and ribbons are the simultaneous band.

### 6.3 Aggregations

With cohort shares  $\pi_g = P(G = g \mid G \neq \infty)$ , the four summaries of `aggregate_att()` are

| Type     | Parameter                                                 | Question                                       |
|----------|-----------------------------------------------------------|------------------------------------------------|
| dynamic  | $\theta_D(e) = \sum_g \pi_{g e} ATT(g, g + e)$            | effect $e$ periods after adoption              |
| group    | $\theta_S(g) = \frac{1}{T-g+1} \sum_{t \geq g} ATT(g, t)$ | average effect experienced by cohort $g$       |
| calendar | $\theta_C(t) = \sum_{g \leq t} \pi_{g t} ATT(g, t)$       | average effect in period $t$ among the treated |
| simple   | $\sum_{g \leq t} \pi_g ATT(g, t) / \sum_{g \leq t} \pi_g$ | one number, overweighting early cohorts        |

The dynamic aggregate at  $e$  averages only the cohorts observed  $e$  periods after adoption. Its composition therefore changes with  $e$ , and the curve can bend for that reason alone. `balance_e = e_0` keeps only cohorts observed for at least  $e_0$  periods, so every point averages the same cohorts.

```
agg_dyn <- aggregate_att(cs_notyet, type = "dynamic", min_e = -4, max_e = 5)
agg_grp <- aggregate_att(cs_notyet, type = "group")
agg_cal <- aggregate_att(cs_notyet, type = "calendar")
agg_simple <- aggregate_att(cs_notyet, type = "simple")
rbind(dynamic = agg_dyn$overall, group = agg_grp$overall, calendar = agg_cal$overall, simple = agg_simple$overall)
#>      estimate std.error  conf.low  conf.high
#> dynamic    7.08269  0.1148962   6.857498   7.307883
#> group     15.04325  0.3176347  14.420695  15.665800
#> calendar  16.23729  0.3068489  15.635882  16.838708
#> simple    18.37981  0.3859830  17.623299  19.136325
agg_grp$by[, c("group", "estimate", "std.error")]
```

```
#>   group estimate std.error
#> 1  1986 28.462138 0.08989571
#> 2  1992 12.964421 0.08240993
#> 3  1998  3.551433 0.10780682
```

The four overall numbers answer four questions. The group aggregate weights cohorts by size; the simple aggregate weights *cells* by cohort size, so cohorts treated early, with many post periods, dominate. Report the one that matches the question, and say which it is.

## 6.4 Sun and Abraham

Sun and Abraham (2021) fix the dynamic regression rather than abandon it: interact the event-time dummies with cohort dummies, use the never-treated (or last-treated) cohort as the only comparison, estimate the saturated regression, and aggregate the cohort-specific coefficients with cohort shares. Without covariates this is numerically the Callaway-Sant’Anna estimator with the never-treated comparison and the universal base period. `fixest::sunab()` does the interaction and the aggregation.

```
sa1 <- feols(y ~ unab(g, year) | id + year, data = sd1, cluster = ~id)
cs_uni <- att_gt(sd1, id = "id", time = "year", group = "g", y = "y", control_group = "never",
  base_period = "universal", n_boot = 0)
event_study_frame("Sun-Abraham" = sa1, "CS never, universal" = aggregate_att(cs_uni, type = "dynamic", n_boot = 0)) %>%
  filter(event_time %in% c(-3, 0, 2, 4)) %>% select(method, event_time, estimate) %>%
  tidyr::pivot_wider(names_from = method, values_from = estimate) %>% as.data.frame()
#>   event_time Sun-Abraham CS never, universal
#> 1         -3  0.007433304      0.007433304
#> 2          0  1.983342265      1.983342265
#> 3          2  6.009719787      6.009719787
#> 4          4 10.080004771     10.080004771
```

## 6.5 Imputation estimators

Borusyak, Jaravel, and Spiess (2024) and Gardner (2022) take the model seriously in the untreated state only. Fit  $Y_{it} = \alpha_i + \lambda_t + \varepsilon_{it}$  on untreated observations, never-treated units and not-yet-treated periods, impute  $\hat{Y}_{it}(0)$  for every treated observation, and average  $Y_{it} - \hat{Y}_{it}(0)$  by event time or cohort. Wooldridge (2021) obtains the same estimates from one regression with cohort-by-period interactions.

The estimator uses every pre-treatment period as the baseline, not only  $g - 1$ . That is efficient when parallel trends holds in all periods and the errors are homoskedastic and serially uncorrelated, and it is the reason its standard errors are often smaller. The price is a stronger assumption, parallel trends in all periods, and a separate pre-trends check, run on the untreated sample with lead dummies.

```
imp1 <- did_imputation(sd1, id = "id", time = "year", group = "g", y = "y", horizons = 0:5, pre_window = 3)
imp1
#> Imputation difference-in-differences (Borusyak, Jaravel, and Spiess)
#> Treated observations: 4182; untreated observations used in the first stage: 15492; standard errors clustered by id
#> Overall ATT: 7.1 (SE 0.09998)
#>   event_time estimate std.error n_cells conf.low conf.high
#>   0         2.033  0.05964    697    1.916    2.150
#>   1         4.110  0.07750    697    3.958    4.262
#>   2         6.027  0.09882    697    5.833    6.221
#>   3         8.187  0.12245    697    7.947    8.427
#>   4        10.106  0.14994    697    9.812   10.400
#>   5        12.137  0.17600    697   11.792   12.482
#> Pre-treatment leads (untreated sample):
#>   event_time estimate std.error conf.low conf.high
#>   -3 -0.002999  0.05475 -0.11030  0.1043
#>   -2  0.069984  0.05749 -0.04270  0.1827
#>   -1  0.026930  0.05462 -0.08013  0.1340
```

## 6.6 Stacked DiD and local-projections DiD

Two regression-based alternatives are common in applied work.

**Stacked DiD** (Cengiz et al. 2019) builds one clean data set per cohort: the cohort, the never-treated units, and the units not yet treated within an event window, then stacks the data sets and runs a TWFE event study with data-set-specific unit and period effects. Every comparison is clean by construction. The implicit weights across cohorts are those of least squares on the stacked sample, roughly proportional to cohort size.

```
stack <- bind_rows(lapply(c(1986, 1992, 1998), function(g0) {
  sd1 %>% filter(g == g0 | g == 0 | g > g0 + 5, year >= g0 - 5, year <= g0 + 5) %>%
    mutate(stack_id = g0, rel = ifelse(g == g0, year - g0, -1000))
}))
stacked <- feols(y ~ i(rel, ref = c(-1, -1000)) | id~stack_id + year~stack_id, data = stack, cluster = ~id)
```

**Local-projections DiD** (Dube, Girardi, Jorda, and Taylor 2023) runs, for each horizon  $h$ , a regression of  $Y_{i,t+h} - Y_{i,t-1}$  on an indicator for adopting at  $t$ , using only clean controls (units not yet treated by  $t + h$ ). It is a sequence of long-difference two-by-twos, one per horizon, and delivers the same logic as the cell-based estimator in regression form.

## 6.7 Comparing estimators

```
imp1_frame <- event_study_frame("Imputation" = imp1) %>% filter(event_time >= 0)
cmp_frame <- event_study_frame("Callaway-Sant'Anna" = agg_dyn, "Sun-Abraham" = sa1,
  "Stacked" = stacked, "Imputation" = imp1_frame) %>%
  filter(event_time >= -4, event_time <= 5)
plot_event_study(cmp_frame) +
  geom_point(data = truth_sd1, aes(x = event_time, y = att), inherit.aes = FALSE, shape = 4, size = 3)
```

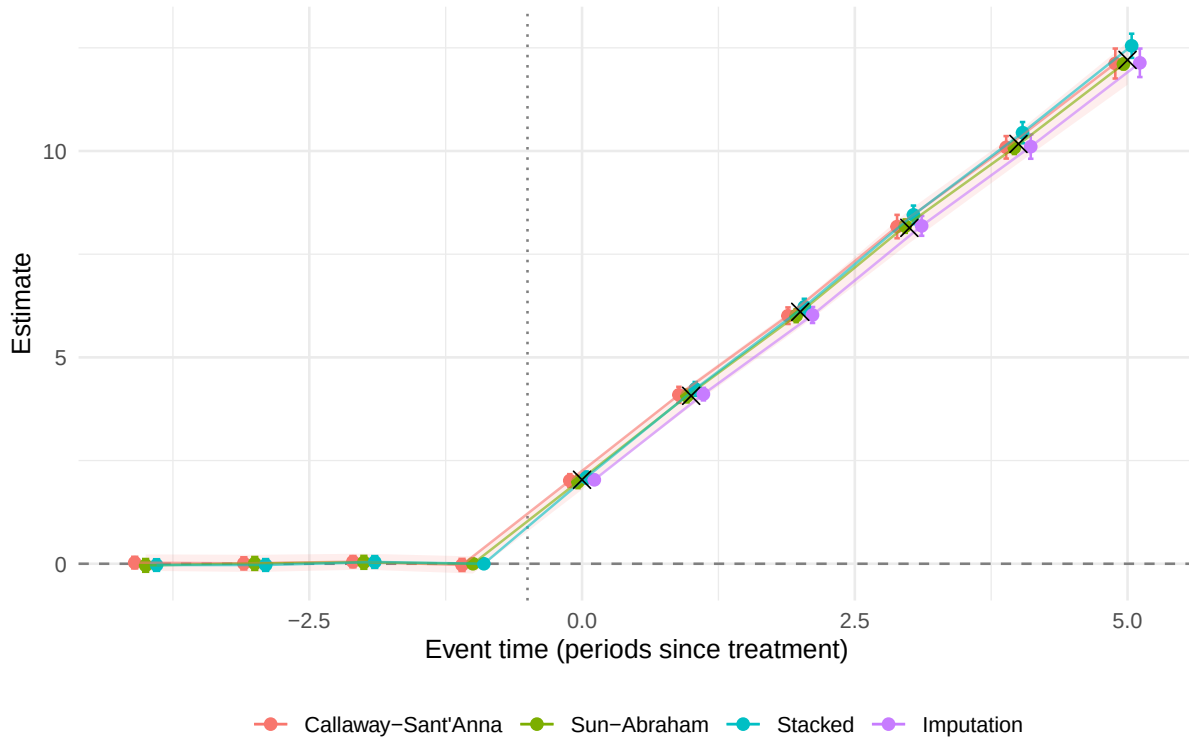


Figure 3: Four heterogeneity-robust estimators on one draw of the staggered design, with the truth as crosses.

```
cmp_frame %>% filter(event_time %in% c(-2, 0, 3, 5)) %>% select(method, event_time, estimate) %>%
  tidyr::pivot_wider(names_from = method, values_from = estimate) %>%
  left_join(truth_sd1, by = "event_time") %>% as.data.frame()
#>   event_time Callaway-Sant'Anna Sun-Abraham Stacked Imputation att
#> 1         -2      0.05018534  0.03444465  0.04305351      NA      NA
#> 2          0      2.01589419  1.98334227  2.08986946  2.033081  2.034433
```

```
#> 3      3      8.16942521 8.17986983 8.45294152 8.186612 8.137733
#> 4      5      12.11951526 12.10724308 12.54894826 12.136702 12.206600
```

| Estimator                             | Parallel trends used          | Comparison group                        | Covariates                               | Precision                                       |
|---------------------------------------|-------------------------------|-----------------------------------------|------------------------------------------|-------------------------------------------------|
| Callaway-Sant'Anna                    | post-treatment periods only   | never or not yet treated                | regression, IPW, doubly robust, per cell | baseline                                        |
| Sun-Abraham                           | post-treatment periods only   | never or last treated                   | linear in the regression                 | equals CS without covariates                    |
| Imputation (BJS, Gardner, Wooldridge) | all periods                   | all untreated observations              | linear in the first stage                | higher under homoskedastic, uncorrelated errors |
| Stacked                               | all periods within the window | never and not yet treated in the window | linear in the regression                 | depends on the window                           |

The estimators agree when their assumptions hold. They disagree when parallel trends holds only after treatment, when pre-periods are contaminated by anticipation, or when covariates matter, and then the choice is about assumptions, not about software.

## 6.8 Covariates in staggered designs

Covariates enter `att_gt()` cell by cell, with a propensity score and an outcome regression fitted on each cell's sample, so the conditional parallel trends assumption of Section 3 is used at the cell level. The regression-based estimators can include covariates only linearly, and a time-invariant covariate must be interacted with time to matter at all.

```
sdx <- sim_did_panel(n_units = 800, n_periods = 9, groups = c(4, 7), never_share = 0.4,
                    mu = c(2, 1), x_trend = 2, x_select = 2, seed = 13)
sdx$x1t <- sdx$x1 * sdx$time
cs_x <- aggregate_att(att_gt(sdx, id = "id", time = "time", group = "g", y = "y", x = "x1",
                             control_group = "notyet", n_boot = 0), type = "dynamic", n_boot = 0)
truth_x <- attr(sdx, "att_gt") %>% mutate(event_time = time - group) %>%
  inner_join(cs_x$weights, by = c("event_time", "group")) %>%
  group_by(event_time) %>% summarise(truth = sum(weight * att), .groups = "drop")
cs_nox <- aggregate_att(att_gt(sdx, id = "id", time = "time", group = "g", y = "y",
                             control_group = "notyet", n_boot = 0), type = "dynamic", n_boot = 0)
sa_x <- feols(y ~ sunab(g, time) + x1t | id + time, data = sdx, cluster = ~id)
twfe_x <- feols(y ~ treated + x1t | id + time, data = sdx, cluster = ~id)
c(static_twfe_plus_x1t = coef(twfe_x)[["treated"]])
#> static_twfe_plus_x1t
#>      3.17151
event_study_frame("CS with x1" = cs_x, "CS without x1" = cs_nox, "Sun-Abraham + x1*t" = sa_x) %>%
  filter(event_time %in% 0:2) %>% select(method, event_time, estimate) %>%
  tidyr::pivot_wider(names_from = method, values_from = estimate) %>%
  left_join(truth_x, by = "event_time") %>% as.data.frame()
#>   event_time CS with x1 CS without x1 Sun-Abraham + x1*t truth
#> 1      0      1.593563      1.803717      1.592278 1.468085
#> 2      1      2.943313      3.376607      2.920277 2.936170
#> 3      2      4.414654      5.137209      4.449403 4.404255
```

The static regression with the covariate trend is far from every true effect. The cell estimator without the covariate is biased upward, because early adopters have higher `x1` and hence steeper untreated trends; with the covariate it is centred. Sun-Abraham with `x1 * t` also works here, but only because the covariate trend is linear in the design.

## 7 Assessing parallel trends

### 7.1 Pre-trends tests and what they cannot tell

A pre-trends plot tests parallel trends *before* treatment. The identifying assumption is about trends *after* treatment. The test is informative only through an extra step: if trends were parallel before, they are probably parallel after. That step is economics, not statistics, and it fails when treatment is timed in response to a change in the outcome, which is the usual reason policies are adopted.

Even on its own terms the test is weak. Roth (2022) shows that pre-trends tests have low power against violations large enough to matter. `pretrend_power()` makes this concrete: it takes the event-study estimates and their covariance, posits a linear violation of parallel trends with a given slope, and reports the probability that the standard pre-test (any pre-period coefficient significant at 5 percent) rejects.

```
pp <- pretrend_power(agg_dyn, n_sim = 3000, seed = 1)
pp$detectable
#> target_power      slope
#> 1          0.5 0.03539882
#> 2          0.8 0.05177628
```

A violation that the test detects only half the time would bias the effect at  $e = 4$  by five times that slope. In this design the effects are large relative to such slopes; in many applications they are not.

### 7.2 Conditioning on passing the pre-test

Worse, the estimates that survive the test are not a random sample of the estimates. When a real violation exists, the draws that fail to reject are those in which the noise happened to offset the pre-trend, and the same noise pushes the post-treatment estimates away from the truth. The bias conditional on passing exceeds the unconditional bias.

```
pp50 <- pretrend_power(agg_dyn, slope = pp$detectable$slope[1], n_sim = 3000, seed = 1)
cbind(power = round(pp50$slopes[[1]]$power, 2), pp50$slopes[[1]]$bias)
#> power event_time unconditional_bias conditional_bias
#> 1 0.5 0 0.03539882 0.03510892
#> 2 0.5 1 0.07079763 0.07323852
#> 3 0.5 2 0.10619645 0.11139009
#> 4 0.5 3 0.14159527 0.14668514
#> 5 0.5 4 0.17699409 0.18009811
#> 6 0.5 5 0.21239290 0.21684166
```

The practical conclusion is not to abandon pre-trends plots, but to stop treating a passed test as a license. Report the plot, report the power of the test, and quantify how conclusions change under plausible violations.

### 7.3 Sensitivity analysis

Rambachan and Roth (2023) replace “parallel trends holds” by a bound on how far it can fail, indexed by the pre-treatment evidence. Two families of bounds are common.

- **Relative magnitudes**  $\Delta^{RM}(\bar{M})$ : the post-treatment violation in any period is at most  $\bar{M}$  times the largest pre-treatment violation.  $\bar{M} = 1$  says trends after treatment deviate no more than they did before.
- **Smoothness**  $\Delta^{SD}(M)$ : the violation evolves smoothly; the change in its slope between consecutive periods is at most  $M$ .  $M = 0$  allows a linear pre-trend to continue.

For each bound the method returns a confidence set for the effect that is valid for every violation in the set. The **breakdown value** is the bound at which zero enters the set. `honestdid_inputs()` extracts the coefficients and their covariance from a dynamic aggregate estimated with the universal base period, so the reference period is exactly zero.

```
csu_h <- att_gt(sdl, id = "id", time = "year", group = "g", y = "y", control_group = "notyet",
               base_period = "universal", n_boot = 0)
dyn_h <- aggregate_att(csu_h, type = "dynamic", min_e = -4, max_e = 3, n_boot = 0)
```

```

hi <- honestdid_inputs(dyn_h)
c(numPrePeriods = hi$numPrePeriods, numPostPeriods = hi$numPostPeriods)
#>   numPrePeriods numPostPeriods
#>           3           4
l1 <- HonestDiD::basisVector(1, hi$numPostPeriods)
orig <- HonestDiD::constructOriginalCS(betahat = hi$betahat, sigma = hi$sigma, numPrePeriods = hi$numPrePeriods,
                                     numPostPeriods = hi$numPostPeriods, l_vec = l1)
rm <- HonestDiD::createSensitivityResults_relativeMagnitudes(
  betahat = hi$betahat, sigma = hi$sigma, numPrePeriods = hi$numPrePeriods, numPostPeriods = hi$numPostPeriods,
  Mbarvec = c(0.5, 1, 2), l_vec = l1, grid.lb = orig$lb - 2, grid.ub = orig$ub + 2)
sm <- HonestDiD::createSensitivityResults(
  betahat = hi$betahat, sigma = hi$sigma, numPrePeriods = hi$numPrePeriods, numPostPeriods = hi$numPostPeriods,
  Mvec = c(0, 0.05, 0.1), l_vec = l1)
rbind(as.data.frame(orig)[, c("lb", "ub", "method")], as.data.frame(rm)[, c("lb", "ub", "method")],
      as.data.frame(sm)[, c("lb", "ub", "method")])
#>      lb      ub  method
#> 1 1.866603 2.165185 Original
#> 2 1.845930 2.203070    C-LF
#> 3 1.794296 2.254704    C-LF
#> 4 1.643695 2.401002    C-LF
#> 5 1.821972 2.150075    FLCI
#> 6 1.769239 2.240533    FLCI
#> 7 1.718311 2.327633    FLCI

```

The grid for the relative-magnitudes set is passed explicitly, because the default grid is built for coefficients on the scale of the package's examples and can miss the estimate entirely.

The rows show the effect at  $e = 0$  under the original assumption, under relative-magnitude violations up to half, once, and twice the largest pre-treatment deviation, and under smoothness bounds of 0, 0.05, and 0.1 per period. The estimated pre-trends are small here, so even  $\bar{M} = 2$  leaves the conclusion intact. When they are not small, this table is the honest statement of what the design can deliver.

## 7.4 Functional-form test

Section 2.5 showed that parallel trends on one scale does not imply it on another. Roth and Sant'Anna (2023) give a test of the stronger claim that it holds on every scale, which is the same as parallel trends of the whole distribution. Under that claim the counterfactual distribution of treated outcomes after treatment is  $F_{1,\text{pre}} + F_{0,\text{post}} - F_{0,\text{pre}}$ , which must be a distribution function: its implied density must be non-negative everywhere. A negative implied mass anywhere rejects distributional parallel trends, and with it the hope that the scale does not matter.

```

breaks <- quantile(c(y1, y2), probs = seq(0, 1, 0.1))
mass <- function(v) as.numeric(table(cut(v, breaks, include.lowest = TRUE))) / length(v)
implied <- mass(y1[treat == 1]) + mass(y2[treat == 0]) - mass(y1[treat == 0])
round(implied, 3)
#> [1] -0.385 -0.262 0.117 0.278 0.291 0.279 0.236 0.166 0.158 0.122

```

The negative entries say that no monotone transformation of this outcome satisfies parallel trends, so the level and log analyses cannot both be right, which is what Section 2.5 found.

## 7.5 Placebos and design checks

- **Placebo outcomes:** an outcome the policy cannot affect should show no effect; if it does, the comparison group is trending differently.
- **Placebo timing:** pretend treatment started earlier and estimate on the pre-period only; the cells of `att_gt()` with `post = 0` are this check, done period by period.
- **Anticipation:** date treatment at the announcement or use `anticipation = k`; a jump at  $e = -1$  is a symptom (Section 4.2).
- **Composition:** in repeated cross-sections, test that the distribution of covariates within groups does not change over time.

- **Estimator agreement:** Callaway-Sant’Anna with never and with not-yet comparisons, Sun-Abraham, and imputation should agree when parallel trends holds in all periods; a gap between them is evidence about the pre-period assumption, not a choice of software.

## 8 Inference

### 8.1 What level to cluster

Two views justify clustering. In the **sampling** view, units are drawn from a population in clusters, and standard errors must reflect the cluster draw: at least the unit, because a unit’s periods are one draw, and higher when units are sampled by state or firm. In the **design** view, treatment is assigned at some level, and the uncertainty is about which clusters were assigned; standard errors should then be clustered at the assignment level. Both views lead to the same practical rule: cluster at the level of treatment assignment, and never below the unit.

```
agg_unit <- aggregate_att(cs_notyet, type = "simple", n_boot = 199, seed = 1)
cs_state <- att_gt(sdl, id = "id", time = "year", group = "g", y = "y", control_group = "notyet",
  cluster = "state", n_boot = 199, seed = 1)
agg_state <- aggregate_att(cs_state, type = "simple")
rbind(cluster_unit = agg_unit$overall, cluster_state = agg_state$overall)
#>               estimate std.error conf.low conf.high
#> cluster_unit  18.37981  0.3690561 17.65648  19.10315
#> cluster_state 18.37981  1.6720317 15.10269  21.65693
```

Treatment is assigned by state, so the state-clustered standard error is the relevant one; it is several times larger than the unit-clustered one.

### 8.2 Few clusters

Cluster-robust standard errors rely on a central limit theorem across clusters. With few clusters, and especially few *treated* clusters, the approximation fails, standard errors are biased downward, and tests over-reject. The wild cluster bootstrap repairs moderate cases but not the case of one or two treated clusters. Randomization inference asks a different question: if treatment timing had been assigned at random across clusters, how unusual is the observed statistic? Under random timing this test is exact for the sharp null of no effect for any unit, and with a studentized statistic it is also asymptotically valid for the weak null of a zero average effect (Roth and Sant’Anna 2023).

```
few <- sim_did_panel(n_units = 300, n_periods = 8, groups = 5, never_share = 0.75, mu = 1, seed = 14)
few$state <- (few$id - 1) %/% 15 + 1
few$g <- ave(few$g, few$state, FUN = function(v) v[1])
few$treated <- as.integer(few$g > 0 & few$time >= few$g)
c(treated_states = length(unique(few$state[few$g > 0])), states = length(unique(few$state)))
#> treated_states      states
#>           4         20
asy <- feols(y ~ treated | id + time, data = few, cluster = ~state)
perm <- did_permutation_test(few, id = "id", time = "time", group = "g", y = "y",
  cluster = "state", n_perm = 499, seed = 1)
c(asymptotic_p = fixest::pvalue(asy)[["treated"]], permutation_p = perm$p.value)
#> asymptotic_p permutation_p
#>    0.0636749    0.1420000
```

### 8.3 Design-based inference and random timing

When treatment timing really is random, as in a randomized rollout, parallel trends is implied but the DiD estimator is not efficient. Roth and Sant’Anna (2023) show that the efficient estimator in a large class subtracts from the simple post-treatment comparison a regression adjustment on pre-treatment differences, in the spirit of covariate adjustment in experiments, and that standard-error reductions of a factor of two are common. They also provide the studentized permutation test used above. `staggered_efficient()` implements the estimator (reproducing the `staggered` package): `beta = 1` is the Callaway-Sant’Anna comparison, `beta = 0` the difference in means, and the default is the plug-in efficient `beta*`. The gain is largest



when pre-treatment outcomes predict post-treatment outcomes weakly, so that subtracting the pre-period difference adds noise rather than removing it.

```
set.seed(19)
n_rt <- 600; T_rt <- 8
g_rt <- sample(c(3, 5, 7, Inf), n_rt, replace = TRUE) # random timing
y_rt <- matrix(rnorm(n_rt * T_rt), n_rt) + outer(rep(1, n_rt), seq_len(T_rt) / 4) # little serial correlation
for (i in seq_len(n_rt)) if (is.finite(g_rt[i])) y_rt[i, seq_len(T_rt) >= g_rt[i]] <- y_rt[i, seq_len(T_rt) >= g_rt[i]] + 1
rt <- data.frame(id = rep(seq_len(n_rt), each = T_rt), time = rep(seq_len(T_rt), n_rt),
                 g = rep(g_rt, each = T_rt), y = as.vector(t(y_rt)))

rbind(
  efficient = staggered_efficient(rt, id = "id", time = "time", group = "g", y = "y", estimand = "simple",
                                n_perm = 199, seed = 1)$estimates[, c("estimate", "std.error", "beta", "fisher_p.value")],
  did_beta_1 = staggered_efficient(rt, id = "id", time = "time", group = "g", y = "y", estimand = "simple",
                                beta = 1, n_perm = 199, seed = 1)$estimates[, c("estimate", "std.error", "beta", "fisher_p.value")]
)

#>           estimate std.error      beta fisher_p.value
#> efficient  0.9564997 0.03859022 -0.03913654          0
#> did_beta_1 0.9744222 0.07369703  1.00000000          0
```

With independent noise across periods the efficient `beta*` is near zero, and the DiD standard error is about 1.4 times larger, the square root of two that differencing costs. Random timing is a stronger assumption than parallel trends; under it the gain in precision is free, and without it the DiD comparison remains the safe choice.

## 9 Beyond binary absorbing treatments

### 9.1 Continuous and multi-valued doses

Many treatments come in doses: a minimum wage increase of 5 or 15 percent, an ad budget, a tax rate. With a dose  $D \geq 0$  and untreated units at  $D = 0$ , two questions arise, and Callaway, Goodman-Bacon, and Sant’Anna (2024) show they need different assumptions.

- **Level effects**  $ATT(d | d) = \mathbb{E}[Y_2(d) - Y_2(0) | D = d]$ : the effect of dose  $d$  for units that received  $d$ . Parallel trends between dose- $d$  units and untreated units identifies it by the usual DiD,  $\mathbb{E}[\Delta Y | D = d] - \mathbb{E}[\Delta Y | D = 0]$ .
- **Causal responses**  $ACRT(d | d) = \partial \mathbb{E}[Y_2(l) | D = d] / \partial l$  at  $l = d$ : the effect of a marginal increase in the dose. Comparing  $ATT(d | d)$  across doses does not identify it, because units that chose different doses may have different responses; the comparison mixes the causal response with selection into dose. Identification needs **strong parallel trends**: every dose group would have followed the same average path at every dose, which rules out selection on gains.

The two-way fixed effects coefficient on a continuous dose is  $\text{Cov}(\Delta Y, D) / \text{Var}(D)$ . It has two readings. It is a weighted sum of level effects with weights that can be negative, and it is a weighted sum of causal-response increments with non-negative weights that integrate to one. The second reading is the appealing one and is the one that needs strong parallel trends. `att_dose()` reports both weight sets so the reader sees which assumption is being invoked.

```
set.seed(15)
nd <- 1500
dose <- ifelse(runif(nd) < 0.3, 0, runif(nd, 0.5, 3))
dd <- data.frame(id = rep(seq_len(nd), each = 2), time = rep(1:2, nd), dose = rep(dose, each = 2))
dd$y <- rep(rnorm(nd), each = 2) + 0.5 * dd$time +
  (dd$time == 2) * (2 * dd$dose - 0.3 * dd$dose^2) + rnorm(2 * nd, sd = 0.5)
ad <- att_dose(dd, id = "id", time = "time", y = "y", dose = "dose", dose_type = "spline",
              n_bins = 4, n_boot = 199, seed = 1)
ad$by_dose[, c("bin", "dose", "att", "std.error")]
#>      bin      dose      att std.error
#> 1 [0.503,1.23] 0.8545172 1.502484 0.06104677
#> 2 (1.23,1.82] 1.5316649 2.283299 0.05646244
#> 3 (1.82,2.42] 2.1181336 2.921821 0.05557766
#> 4 (2.42,3] 2.7044865 3.232230 0.05564971
ad$twfe_weights
```



```
#>      dose      share    w_level  w_slope
#> 1 0.000000 0.2806667 -0.36569599      NA
#> 2 0.8545172 0.1800000 -0.07991391 0.3656960
#> 3 1.5316649 0.1800000 0.04261047 0.4456099
#> 4 2.1181336 0.1793333 0.14817640 0.4029994
#> 5 2.7044865 0.1800000 0.25482303 0.2548230

plot_att_dose(ad, "acr")
```

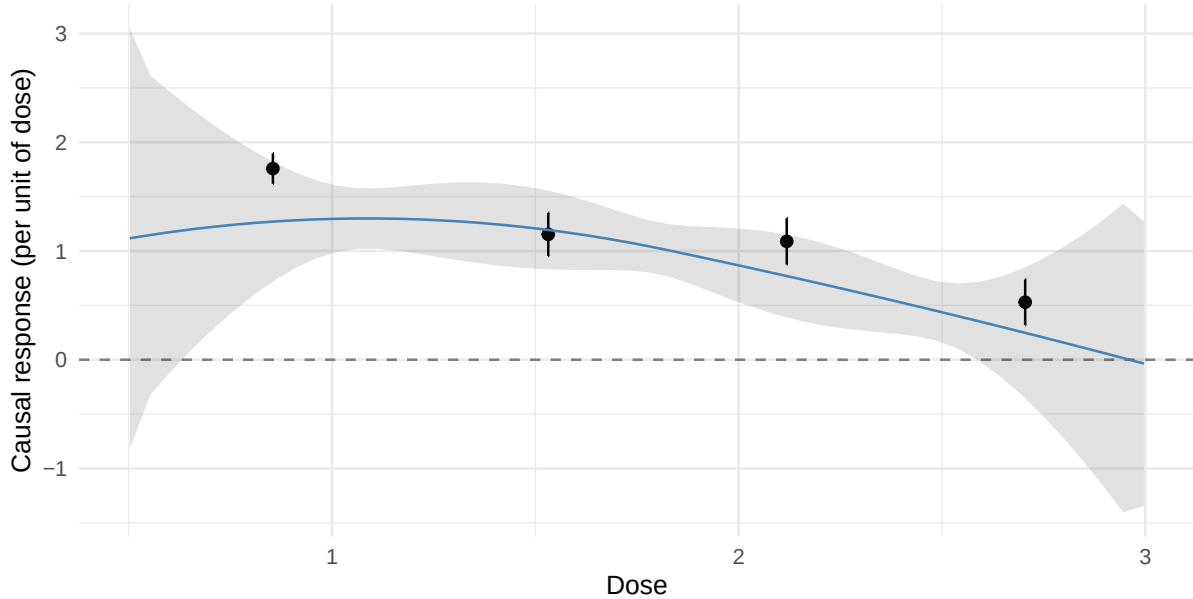


Figure 4: Causal responses by dose bin (points) and the spline estimate with a uniform band; the true response is  $2 - 0.6d$ .

In the design the response curve is  $2d - 0.3d^2$ , so the true causal response is  $2 - 0.6d$  and falls with the dose. The level weights put negative mass on low doses; the slope weights are positive and sum to one over the dose gaps. A researcher who reports the TWFE coefficient as “the effect of a unit of dose” is using the second reading and therefore strong parallel trends, whether or not they say so.

## 9.2 Treatments that turn on and off

When treatment can be switched off and on, a unit’s history is a sequence  $(d_1, \dots, d_T)$ , and there are  $2^{T-1}$  possible groups instead of  $T$ . The building block becomes  $ATT(\mathbf{d}, t)$ , the effect of sequence  $\mathbf{d}$  versus never being treated. Identification works as in Section 2.4 once groups are defined by the first exposure date and parallel trends is assumed across all sequences. Mechanically, `att_gt()` with the first exposure period as the group variable computes the right cells. The difficulty is interpretation: the cell for a first-exposure cohort at time  $t$  averages units that stayed treated, switched off, and switched back on. de Chaisemartin and D’Haultfoeuille (2024) propose normalizing the event-study aggregate by the average number of treated periods, which reads as an effect “per unit of treatment” only under no carryover. With rich switching, the honest conclusion is often that DiD is not the right tool, and that sequential methods for time-varying treatments (marginal structural models, panel matching) fit better.

## 9.3 Triple differences

A triple difference adds a third dimension: within each group and period, some units are eligible for the policy and others are not (Gruber 1994). The estimand subtracts the DiD among ineligible units from the DiD among eligible units. It identifies the ATT under a weaker assumption than either DiD alone:

the *difference* in untreated trends between eligible and ineligible units must be the same in the treated and comparison groups. Without covariates the estimate is the coefficient on the triple interaction, or equivalently the difference of two `att_gt()` cells. With covariates the shortcut fails: Ortiz-Villavicencio and Sant’Anna (2025) show that neither the triple-interaction regression with controls nor the difference of two conditional DiDs recovers the ATT, and they give a doubly robust estimator built directly from the triple-difference identification. The same holds for staggered adoption.

```
set.seed(16)
n3 <- 2000
d3 <- data.frame(id = rep(seq_len(n3), each = 2), time = rep(1:2, n3),
                 treat = rep(rbinom(n3, 1, 0.5), each = 2), elig = rep(rbinom(n3, 1, 0.5), each = 2))
d3$post <- as.integer(d3$time == 2)
d3$y <- rnorm(2 * n3) + 0.4 * d3$post * d3$treat + 0.3 * d3$post * d3$elig +
  1.5 * d3$post * d3$treat * d3$elig # effect 1.5 on eligible units in the treated group
ddd <- feols(y ~ treat:post:elig + treat:post + post:elig | id + time, data = d3, cluster = ~id)
did_two <- function(e) coef(feols(y ~ treat:post | id + time, data = subset(d3, elig == e)))[["treat:post"]]
c(triple_interaction = coef(ddd)[["treat:post:elig"]], difference_of_dids = did_two(1) - did_two(0))
#> triple_interaction difference_of_dids
#> 1.4478 1.4478
```

## 9.4 Nonlinear and distributional DiD

Section 2.5 showed that the scale is part of the assumption. Three situations call for models beyond the linear DiD.

- **Counts and positive outcomes with zeros.** Logs are undefined at zero and the log DiD has no clean interpretation. Wooldridge (2023) proposes a Poisson quasi-likelihood with unit and period effects, whose identifying assumption is parallel trends in *ratios*: the proportional change in untreated means is the same across groups. The coefficient gives the proportional ATT as  $\exp(\beta) - 1$ , and `fixest::fepois()` estimates it with the same syntax as the linear model. This is the workhorse of the gravity literature (Section 11).
- **Distributional effects.** Changes-in-changes (Athey and Imbens 2006) assumes that the untreated outcome distribution changes over time through one monotone map common to both groups, estimates the map from the comparison group’s pre and post quantiles, and applies it to the treated group’s pre-period distribution. It identifies the entire counterfactual distribution, so quantile effects follow, and it is invariant to monotone transformations by construction. Callaway and Li (2019) give quantile treatment effects on the treated under a copula-type assumption on the joint distribution of changes.

```
set.seed(17)
np <- 3000
pd <- data.frame(id = rep(seq_len(np), each = 2), time = rep(1:2, np), treat = rep(rbinom(np, 1, 0.5), each = 2))
pd$post <- as.integer(pd$time == 2)
mu <- exp(rep(rnorm(np, sd = 0.5), each = 2) + 0.2 * pd$post + 0.3 * pd$post * pd$treat)
pd$y <- rpois(2 * np, mu)
c(proportional_att_poisson = exp(coef(fepois(y ~ treat:post | id + time, data = pd))["treat:post"]) - 1,
  truth = exp(0.3) - 1,
  level_did_over_baseline = coef(feols(y ~ treat:post | id + time, data = pd))["treat:post"] /
    mean(pd$y[pd$treat == 1 & pd$post == 0]))
#> proportional_att_poisson truth level_did_over_baseline
#> 0.3746572 0.3498588 0.4464655
```

## 10 Synthetic difference-in-differences

### 10.1 When the comparison group is not credible

DiD averages all untreated units into one comparison group. With a handful of aggregate units, say one treated state and forty others, that average has no reason to trend like the treated state. Synthetic control (Abadie, Diamond, and Hainmueller 2010) replaces the average by a weighted combination of untreated units chosen to reproduce the treated unit’s pre-treatment path. Synthetic DiD (Arkhangelsky et al. 2021)

combines the two ideas: it keeps the unit and period effects of DiD, so levels need not match, and it adds unit weights that align the untreated *trends* with the treated trend and period weights that make the pre-treatment periods predictive of the post-treatment ones. The estimator solves

$$(\hat{\tau}, \hat{\alpha}, \hat{\beta}) = \arg \min \sum_i \sum_t (Y_{it} - \alpha_i - \beta_t - \tau D_{it})^2 \hat{\omega}_i \hat{\lambda}_t,$$

a weighted two-way fixed effects regression. Plain DiD is the case of uniform weights; synthetic control is the case of fitted unit weights, no period weights, and no unit effects.

## 10.2 Unit and time weights

The unit weights solve a regularized constrained regression of the treated pre-treatment average on the untreated units' pre-treatment paths, with an intercept, so only trends must match. The period weights solve the analogous problem in the other direction, predicting untreated post-treatment averages from their pre-treatment periods. The regularization  $\zeta$  is set from the noise in first differences, and the weights are sparse in practice. `sdid_weights()` returns them, with a cell-weight column for the regression step in `fixest`.

The design below has a latent factor: units load differently on a common time trend, and treated units load more. Plain DiD compares treated units to an average that trends differently; synthetic DiD reweights the untreated units to match.

```
set.seed(18)
n0 <- 30; n1 <- 5; t0 <- 10; t1 <- 5
nn <- n0 + n1; TT <- t0 + t1
f <- cumsum(rnorm(TT)) + seq_len(TT) / 2
load <- c(runif(n0, 0, 2), runif(n1, 1.5, 2.5))
Y <- outer(rnorm(nn), rep(1, TT)) + outer(load, f) + matrix(rnorm(nn * TT, sd = 0.3), nn)
Y[(n0 + 1):nn, (t0 + 1):TT] <- Y[(n0 + 1):nn, (t0 + 1):TT] + 2
blk <- data.frame(id = rep(seq_len(nn), each = TT), time = rep(seq_len(TT), nn), y = as.vector(t(Y)),
                  d = as.integer(rep(seq_len(nn) > n0, each = TT) & rep(seq_len(TT) > t0, nn)))
ests <- sapply(c("did", "sc", "difp", "sdid"), function(m)
  sdid_weights(blk, id = "id", time = "time", y = "y", d = "d", estimator = m)$estimate)
round(ests, 3)
#>   did   sc  difp  sdid
#> 7.019 3.020 2.896 2.006
sw <- sdid_weights(blk, id = "id", time = "time", y = "y", d = "d")
fit_w <- feols(y ~ d | id + time, data = merge(blk, sw$cell_weights, by = c("id", "time")), weights = ~weight)
c(closed_form = sw$estimate, weighted_fixest = coef(fit_w)[["d"]])
#>      closed_form weighted_fixest
#>      2.006123      2.006123
```

The true effect is 2. DiD is far off because the untreated average has a flatter trend than the treated units. Synthetic control reduces the bias but does not remove it, because without unit effects its weights must match levels as well as trends. Synthetic DiD matches trends only and recovers the effect, and its closed form equals the weighted regression.

## 10.3 Inference

Three resampling schemes are available. The **placebo** scheme draws untreated units, treats them as if treated, and re-estimates; it is the choice when treated units are few, because it uses the untreated units' variability. The **bootstrap** resamples units with replacement and needs several treated units. The **jackknife** leaves one unit out. All three re-solve the weights on each draw, so the standard error includes the uncertainty from weight estimation.

```
sapply(c("placebo", "bootstrap", "jackknife"), function(m)
  sdid_se(sw, method = m, n_reps = 100, seed = 1)$std.error)
#> placebo bootstrap jackknife
#> 0.1257903 0.1251879 0.1476571
```

# 10.4 Diagnostics

plot\_sdid(sw)



Figure 5: Treated average and the weighted untreated average; bars show the period weights.

Read three things from the fit. The pre-treatment paths should coincide after the unit effects are removed; a gap there is a failure of the weights, not evidence of an effect. The unit weights should not concentrate on one or two donors, which makes the estimate fragile to those units. And the period weights show which pre-treatment periods are doing the work; weights on distant periods with different dynamics are a warning. With staggered adoption, run the estimator cohort by cohort and average the results with cohort weights. The companion synthetic control note in this series covers the Abadie method, its placebo tests, and the `synthdid` package in depth.

# 11 Practice

## 11.1 Replications and anchors

The replication notebooks for this section live in the notes folder and are planned for the package’s replication directory for Section 07; they are not yet part of the package. Each anchors one part of the lecture.

| Paper                                                                              | What it anchors                                                         |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| de Chaisemartin and D’Haultfoeuille (2020), newspapers and turnout; union wages    | negative weights and the switchers-versus-stayers estimator (Section 5) |
| Freyaldenhoven, Hansen, and Shapiro (2019); Roth (2022)                            | pre-trends, power, and the pre-test bias (Section 7)                    |
| Roth and Sant’Anna (2023), functional form; Cengiz et al. (2019) minimum wage bins | scale dependence and the distributional test (Sections 2.5, 7.4)        |
| Elenev et al. (2024), staggered health policy with spillovers                      | staggered estimators and the interference assumption (Sections 6, 12)   |
| Nagengast and Yotov (2025), trade agreements in gravity models                     | nonlinear DiD with Poisson fixed effects (Section 9.4)                  |

| Paper                                                                                                                           | What it anchors                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Arkhangelsky et al. (2021), California Proposition 99; Li (2024), forward DiD                                                   | synthetic DiD (Section 10)                                                                               |
| Planned: Callaway and Sant’Anna (2021) minimum wage; Sant’Anna and Zhao (2020); Goodman-Bacon (2021); Rambachan and Roth (2023) | <code>att_gt()</code> validation, the three faces of covariates, the decomposition, sensitivity analysis |

## 11.2 A DiD checklist

Adapted from Sant’Anna’s course.

1. Plot the treatment rollout: who is treated when, and who never.
2. Count units per cohort and per comparison group; small cohorts have noisy cells and small comparison groups have overlap problems.
3. Plot raw average outcomes by cohort over calendar time before any estimation.
4. State the comparison group and the parallel trends assumption in words: who decides treatment, what do they know, what selection is allowed. Date treatment at the announcement if units can anticipate.
5. Estimate the event study with `att_gt()` and no covariates, universal base period, uniform bands. Read the pre-treatment cells, then the post-treatment ones.
6. If unconditional parallel trends is implausible, add pre-treatment covariates with the doubly robust cells, and check overlap. Do not add them linearly to a TWFE regression.
7. Compare never-treated and not-yet-treated comparisons, Sun-Abraham, and imputation. Disagreement is information about the pre-period assumption.
8. Report the power of the pre-test (`pretrend_power()`) and a sensitivity analysis (HonestDiD) rather than a passed test.
9. Cluster at the level of treatment assignment; with few treated clusters, use `did_permutation_test()`.
10. If parallel trends is not credible with any comparison group, consider synthetic DiD, or a different design.

## 12 Extensions

- **Efficient DiD under parallel trends in all periods.** Chen, Sant’Anna, and Xie (2024) show that when the assumption holds before and after treatment, the model is overidentified, and derive the estimator that weights the pre-treatment periods optimally given the serial correlation of outcomes. It nests Callaway-Sant’Anna and the imputation estimator as special cases and dominates both.
- **Heterogeneous effects.** The doubly robust cell score is a pseudo outcome with conditional mean equal to the conditional ATT given covariates. Regressing it on a subset of covariates gives the best linear approximation of  $CATT(g, t, Z)$  (Callaway, Chen, and Sant’Anna 2024), and a meta-learner built on it estimates it flexibly (the DiD analogue of Section 08’s DR-learner).
- **Selection and parallel trends.** Ghanem, Sant’Anna, and Wuthrich (2022) derive the selection mechanisms under which parallel trends holds, and Marx, Tamer, and Tang (2024) show that forward-looking adoption decisions generally violate it. Both give the vocabulary for step 4 of the checklist.
- **Compositional changes.** In repeated cross-sections the stationarity assumption can fail; Sant’Anna and Xu (2023) give estimators that allow the covariate distribution to change over time and a test for it.
- **Spillovers.** When treated units affect their neighbours, the no-interference assumption fails and the comparison group is contaminated. Elenev et al. (2024) and Butts (2023) estimate direct and spillover effects by defining exposure through distance to treated units.
- **Interactive fixed effects and matrix completion.** Parallel trends is the one-factor case of a model with unit loadings on common factors. Interactive fixed effects (Bai 2009; Xu 2017) and matrix completion (Athey et al. 2021) impute untreated outcomes under a low-rank factor structure; synthetic DiD is a weighted special case. Liu, Wang, and Xu

(2024) unify these as counterfactual estimators with placebo and equivalence tests.

- **Dynamic treatment regimes.** With treatments chosen sequentially in response to past outcomes, the target becomes the effect of a treatment path, identified by the g-formula and estimated by sequential regression or dynamic DML rather than DiD (Section 08).

## 13 Appendix

### 13.1 Influence functions for the two-by-two

**Panel.** With  $D_i = \mathbf{1}\{G_i = 2\}$ ,  $\pi = P(D = 1)$ , and  $\Delta Y_i = Y_{i2} - Y_{i1}$ , the estimator is

$$\hat{\theta} = \frac{\mathbb{E}_n[D \Delta Y]}{\mathbb{E}_n[D]} - \frac{\mathbb{E}_n[(1 - D) \Delta Y]}{\mathbb{E}_n[1 - D]}.$$

Each ratio is a smooth function of two sample means. A first-order expansion of  $a_n/b_n$  around  $(a, b)$  gives  $a_n/b_n - a/b = (a_n - a)/b - a(b_n - b)/b^2 + o_p(n^{-1/2})$ . Applied to the first ratio with  $a = \mathbb{E}[D \Delta Y]$  and  $b = \pi$ ,

$$\frac{\mathbb{E}_n[D \Delta Y]}{\mathbb{E}_n[D]} - \mathbb{E}[\Delta Y \mid D = 1] = \mathbb{E}_n \left[ \frac{D}{\pi} (\Delta Y - \mathbb{E}[\Delta Y \mid D = 1]) \right] + o_p(n^{-1/2}),$$

and likewise for the second ratio. Subtracting,  $\hat{\theta} - \theta = \mathbb{E}_n[\xi_i] + o_p(n^{-1/2})$  with

$$\xi_i = \frac{D_i}{\pi} (\Delta Y_i - \mathbb{E}[\Delta Y \mid D = 1]) - \frac{1 - D_i}{1 - \pi} (\Delta Y_i - \mathbb{E}[\Delta Y \mid D = 0]).$$

$\xi_i$  has mean zero and variance  $\mathbb{E}[\xi^2] = \text{Var}(\Delta Y \mid D = 1)/\pi + \text{Var}(\Delta Y \mid D = 0)/(1 - \pi)$ , so  $\sqrt{n}(\hat{\theta} - \theta) \rightarrow N(0, \mathbb{E}[\xi^2])$  by the central limit theorem, and the plug-in  $\hat{\xi}_i$  gives the standard error  $\sqrt{\sum_i \hat{\xi}_i^2/n}$ .

**Repeated cross-sections.** With  $T_i \in \{1, 2\}$  the period of observation and four cells  $(t, d)$ , the estimator is the DiD of four cell means, and the same expansion gives

$$\xi_i = \xi_i^{(2,1)} - \xi_i^{(1,1)} - \xi_i^{(2,0)} + \xi_i^{(1,0)}, \quad \xi_i^{(t,d)} = \frac{\mathbf{1}\{T_i = t, D_i = d\}}{P(T = t, D = d)} (Y_i - \mathbb{E}[Y \mid T = t, D = d]).$$

**Clusters.** If observations are sampled in clusters, replace  $\hat{\xi}_i$  by cluster sums  $\hat{\xi}_c = \sum_{i \in c} \hat{\xi}_i$ ; the variance is  $\sum_c \hat{\xi}_c^2/n^2$ .

**Covariates.** With parametric first steps the influence function of the regression, IPW, and doubly robust estimators includes correction terms for the estimated logit and regression coefficients (Sant’Anna and Zhao 2020, Appendix). `att_gt()` reports these exact influence functions. With cross-fitted learners it reports the plain orthogonal score, whose first-order corrections vanish by the argument of the next subsection.

### 13.2 Orthogonality and remainder of the DR-DiD score

Write the panel doubly robust score with  $\pi = \mathbb{E}[D]$  and nuisances  $\eta = (p, m)$ :

$$\psi(W; \theta, \eta) = \frac{1}{\pi} \left( D - \frac{p(X)(1 - D)}{1 - p(X)} \right) (\Delta Y - m(X)) - \frac{D}{\pi} \theta.$$

Two facts:  $\mathbb{E}[D \mid X] = p_0(X)$ , so  $\mathbb{E}[p_0(X)(1 - D)/(1 - p_0(X)) \mid X] = p_0(X)$ ; and  $\mathbb{E}[\Delta Y - m_0(X) \mid D = 0, X] = 0$ .

*Direction in  $m$ .* Replacing  $m_0$  by  $m_0 + t\Delta_m$  changes the moment by  $-t\mathbb{E}[(D - p_0(1 - D)/(1 - p_0))\Delta_m(X)]/\pi$ . Conditioning on  $X$ , the bracket has mean  $p_0 - p_0 = 0$ . The derivative is zero.

*Direction in  $p$ .* Only the weight  $p(1 - D)/(1 - p)$  depends on  $p$ , with derivative  $(1 - D)\Delta_p(X)/(1 - p_0(X))^2$ . It multiplies  $\Delta Y - m_0(X)$  on untreated units, whose conditional mean is zero. The derivative is zero. The score is Neyman orthogonal.

*Exact remainder.* With fixed  $\hat{p}, \hat{m}$ , write  $\Delta_p = \hat{p} - p_0$  and  $\Delta_m = \hat{m} - m_0$ . Conditioning on  $X$  and using the two facts, every term involving  $\Delta Y - m_0(X)$  on untreated units vanishes, and what remains is a product of the two nuisance errors:

$$\mathbb{E}[\psi(W; \theta_0, \hat{\eta})] = \frac{1}{\pi} \mathbb{E}\left[\frac{\hat{p}(X) - p_0(X)}{1 - \hat{p}(X)} (m_0(X) - \hat{m}(X))\right].$$

It is zero when either nuisance is correct (double robustness), and it is  $o(n^{-1/2})$  when the product of the  $L^2$  errors is, which is the rate condition of Section 04. The Hajek normalization used in `att_gt()` divides the control term by its own mean instead of  $\pi$ ; the difference is  $O_p(n^{-1/2})$  times the remainder and does not affect the argument.

### 13.3 The Bacon decomposition

**Theorem (Goodman-Bacon 2021).** In a balanced panel with  $K$  timing groups and possibly a never-treated group  $U$ , the static TWFE coefficient satisfies  $\hat{\beta} = \sum_k s_{kU} \hat{\beta}_{kU} + \sum_{k < l} (s_{kl}^k \hat{\beta}_{kl}^k + s_{kl}^l \hat{\beta}_{kl}^l)$ , where each  $\hat{\beta}$  is a two-by-two DiD estimate and the weights are non-negative and sum to one. With  $n_k$  the population share of group  $k$ ,  $\bar{D}_k$  the share of periods it spends treated, and  $\hat{V}^D$  the variance of the doubly demeaned treatment indicator,  $s_{kU} = (n_k + n_U)^2 n_{kU} (1 - n_{kU}) \bar{D}_k (1 - \bar{D}_k) / \hat{V}^D$  with  $n_{kU} = n_k / (n_k + n_U)$ , and the two timing weights are given in the paper’s Theorem 1.

*Sketch.* By the Frisch-Waugh-Lovell theorem,  $\hat{\beta} = \text{Cov}(\tilde{D}, Y) / \text{Var}(\tilde{D})$  where  $\tilde{D}$  is  $D$  with unit and period means removed. Restricting the sample to any pair of groups and applying the same formula gives that pair’s two-by-two estimate. The full-sample covariance is a sum over pairs of the within-pair covariances, because  $\tilde{D}$  is piecewise constant across groups and periods; dividing each by  $\text{Var}(\tilde{D})$  yields the weights, which are each pair’s share of the treatment variance. `bacon_decomp()` computes the pieces on group-by-period means, and its `check` component verifies the identity numerically.

### 13.4 Identification of ATT(g,t)

**Never-treated comparison.** Fix  $g$  and  $t \geq g$ . Parallel trends for each period  $s = g, \dots, t$ ,  $\mathbb{E}[Y_s(\infty) - Y_{s-1}(\infty) \mid G = g] = \mathbb{E}[Y_s(\infty) - Y_{s-1}(\infty) \mid G = \infty]$ , summed over  $s$  gives  $\mathbb{E}[Y_t(\infty) - Y_{g-1}(\infty) \mid G = g] = \mathbb{E}[Y_t(\infty) - Y_{g-1}(\infty) \mid G = \infty]$ . No anticipation replaces  $Y_{g-1}(\infty)$  by the observed  $Y_{g-1}$  for group  $g$ , and no interference replaces the remaining potential outcomes by observed ones. Hence  $ATT(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid G = \infty]$ .

**Not-yet-treated comparison.** Replace  $G = \infty$  by the event  $D_t = 0, G \neq g$  in every step; the assumption must then hold for that changing set, which is the not-yet-treated version of Assumption 3.

**Covariates.** With conditional parallel trends and overlap, the same argument conditional on  $X$  identifies  $ATT(g, t \mid X)$ , and averaging over the distribution of  $X$  in group  $g$  gives the outcome regression, IPW, and doubly robust estimands of Section 3.3 with  $\Delta Y$  replaced by  $Y_t - Y_{g-1}$  and  $D$  by  $\mathbf{1}\{G = g\}$  within the cell sample (Callaway and Sant’Anna 2021, Theorem 1).

### 13.5 Multiplier bootstrap and simultaneous bands

Let  $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_K)$  be the cell estimates with  $\sqrt{n}(\hat{\theta} - \theta) = n^{-1/2} \sum_i \xi_i + o_p(1)$  jointly, and let  $V_1, \dots, V_n$  be i.i.d. draws, independent of the data, with mean zero and variance one (Mammen’s two-point distribution or Rademacher). Define  $\hat{\theta}^* = \hat{\theta} + n^{-1} \sum_i V_i \hat{\xi}_i$ . By the conditional multiplier central limit theorem,  $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$  converges conditionally on the data to the same Gaussian limit as  $\sqrt{n}(\hat{\theta} - \theta)$  (Callaway and Sant’Anna 2021, Theorem 3). Therefore:

- the bootstrap standard error, taken as the interquartile range of  $\hat{\theta}_k^* - \hat{\theta}_k$  divided by that of a standard normal, is consistent and robust to outlying draws;

- with  $\hat{c}_{1-\alpha}$  the  $(1 - \alpha)$  quantile across draws of  $\max_k |\hat{\theta}_k^* - \hat{\theta}_k| / \hat{s}e_k$ , the band  $\hat{\theta}_k \pm \hat{c}_{1-\alpha} \hat{s}e_k$  covers all  $K$  cells jointly with asymptotic probability  $1 - \alpha$ .

With clusters, sum  $\hat{\xi}_i$  within clusters and draw one  $V$  per cluster. Aggregations inherit the result because their influence functions are linear combinations of the cell influence functions plus the estimation error of the aggregation weights, which is itself a mean-zero average.

## 13.6 References

- Abadie, Alberto. 2005. “Semiparametric Difference-in-Differences Estimators.” *Review of Economic Studies*.
- Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. “Synthetic Control Methods for Comparative Case Studies.” *Journal of the American Statistical Association*.
- Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. 2021. “Synthetic Difference-in-Differences.” *American Economic Review*.
- Athey, Susan, and Guido W. Imbens. 2006. “Identification and Inference in Nonlinear Difference-in-Differences Models.” *Econometrica*.
- Athey, Susan, and Guido W. Imbens. 2022. “Design-Based Analysis in Difference-in-Differences Settings with Staggered Adoption.” *Journal of Econometrics*.
- Athey, Susan, Mohsen Bayati, Nikolay Doudchenko, Guido Imbens, and Khashayar Khosravi. 2021. “Matrix Completion Methods for Causal Panel Data Models.” *Journal of the American Statistical Association*.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. “Revisiting Event-Study Designs: Robust and Efficient Estimation.” *Review of Economic Studies*.
- Butts, Kyle. 2023. “Difference-in-Differences Estimation with Spatial Spillovers.” Working paper.
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant’Anna. 2024. “Difference-in-Differences with a Continuous Treatment.” Working paper, arXiv:2107.02637.
- Callaway, Brantly, and Tong Li. 2019. “Quantile Treatment Effects in Difference in Differences Models with Panel Data.” *Quantitative Economics*.
- Callaway, Brantly, and Pedro H. C. Sant’Anna. 2021. “Difference-in-Differences with Multiple Time Periods.” *Journal of Econometrics*.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. “The Effect of Minimum Wages on Low-Wage Jobs.” *Quarterly Journal of Economics*.
- Chang, Neng-Chieh. 2020. “Double/Debiased Machine Learning for Difference-in-Differences Models.” *The Econometrics Journal*.
- Chen, Xiaohong, Pedro H. C. Sant’Anna, and Haitian Xie. 2024. “Efficient Difference-in-Differences and Event Study Estimators.” Working paper.
- Chernozhukov, Victor, Christian Hansen, Nathan Kallus, Martin Spindler, and Vasilis Syrgkanis. 2026. *Applied Causal Inference Powered by ML and AI*, chapter 16. Online, version 0.1.2.
- de Chaisemartin, Clement, and Xavier D’Haultfoeuille. 2020. “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects.” *American Economic Review*.
- de Chaisemartin, Clement, and Xavier D’Haultfoeuille. 2024. “Difference-in-Differences Estimators of Intertemporal Treatment Effects.” *Review of Economics and Statistics*.
- Dube, Arindrajit, Daniele Girardi, Oscar Jorda, and Alan M. Taylor. 2023. “A Local Projections Approach to Difference-in-Differences Event Studies.” NBER Working Paper 31184.
- Freyaldenhoven, Simon, Christian Hansen, and Jesse M. Shapiro. 2019. “Pre-Event Trends in the Panel Event-Study Design.” *American Economic Review*.
- Gardner, John. 2022. “Two-Stage Differences in Differences.” Working paper.
- Ghanem, Dalia, Pedro H. C. Sant’Anna, and Kaspar Wuthrich. 2022. “Selection and Parallel Trends.” Working paper, arXiv:2203.09001.
- Goodman-Bacon, Andrew. 2021. “Difference-in-Differences with Variation in Treatment Timing.” *Journal of Econometrics*.
- Gruber, Jonathan. 1994. “The Incidence of Mandated Maternity Benefits.” *American Economic Review*.
- Heckman, James J., Hidehiko Ichimura, and Petra E. Todd. 1997. “Matching as an Econometric



- Evaluation Estimator.” *Review of Economic Studies*.
- Liu, Licheng, Ye Wang, and Yiqing Xu. 2024. “A Practical Guide to Counterfactual Estimators for Causal Inference with Time-Series Cross-Sectional Data.” *American Journal of Political Science*.
  - Marx, Philip, Elie Tamer, and Xun Tang. 2024. “Parallel Trends and Dynamic Choices.” *Journal of Political Economy: Microeconomics*.
  - Ortiz-Villavicencio, Marcelo, and Pedro H. C. Sant’Anna. 2025. “Better Understanding Triple Differences Estimators.” Working paper.
  - Rambachan, Ashesh, and Jonathan Roth. 2023. “A More Credible Approach to Parallel Trends.” *Review of Economic Studies*.
  - Roth, Jonathan. 2022. “Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends.” *American Economic Review: Insights*.
  - Roth, Jonathan, and Pedro H. C. Sant’Anna. 2023. “When Is Parallel Trends Sensitive to Functional Form?” *Econometrica*.
  - Roth, Jonathan, and Pedro H. C. Sant’Anna. 2023. “Efficient Estimation for Staggered Rollout Designs.” *Journal of Political Economy: Microeconomics*.
  - Roth, Jonathan, Pedro H. C. Sant’Anna, Alyssa Bilinski, and John Poe. 2023. “What’s Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature.” *Journal of Econometrics*.
  - Sant’Anna, Pedro H. C. 2025. *Causal Inference Using Difference-in-Differences*. Lecture slides, Emory University.
  - Sant’Anna, Pedro H. C., and Qi Xu. 2023. “Difference-in-Differences with Compositional Changes.” Working paper, arXiv:2304.13925.
  - Sant’Anna, Pedro H. C., and Jun Zhao. 2020. “Doubly Robust Difference-in-Differences Estimators.” *Journal of Econometrics*.
  - Sun, Liyang, and Sarah Abraham. 2021. “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects.” *Journal of Econometrics*.
  - Wooldridge, Jeffrey M. 2021. “Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators.” Working paper.
  - Wooldridge, Jeffrey M. 2023. “Simple Approaches to Nonlinear Difference-in-Differences with Panel Data.” *The Econometrics Journal*.